

Sultanate of Oman

University of Technology and Applied Sciences

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of the requirements for the Advanced Diploma Level

Research Methodology Course

Effectiveness of Remote Work on Software Development Teams

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Abstract

This study looks at the differences between remote work and office work in software development, focusing on employee happiness, productivity, and teamwork. We analyzed a dataset from Kaggle, which includes responses from 5,000 employees, using a quantitative approach. We applied machine learning techniques to find out what factors influence employee preferences and how different work environments affect performance.

Our findings show that remote work provides flexibility and saves time on commuting, but it can also make people feel isolated and affect their physical health. Having a good home office setup and organizing regular social interactions can help reduce these issues. We also suggest that a hybrid work model, which combines remote and in-office work, could be the best option. This way, employees can enjoy the benefits of both working styles. We use six machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes, to predict productivity and stress levels for remote workers.

However, there are some limitations to our study. We used synthetic data, which may not fully represent real-life situations, and there may be biases in how people reported their experiences. Additionally, since our dataset focuses on the IT sector, the results may not apply to other industries. Despite these limitations, our study provides useful insights for companies looking to improve their remote and office work policies.

Keywords: - Remote work - Office work - Software development - Employee happiness - Productivity - Teamwork - Dataset - Kaggle - Quantitative approach - Machine learning - Employee preferences - Work environments - Performance - Home office setup - Social interactions - Hybrid work model - Benefits - Limitations - Synthetic data - IT sector - Remote and office work policies.

Statement of Original Authorship

I hereby declare that this report entitled, "**Effectiveness of Remote Work on Software Development Teams**" is our own original work carried out as part of Research Methodology module assessment except the extent, which are from another research and duly acknowledged. All sources used for this report have been fully and properly cited. It contains no material which to a substantial extent has been accepted or used at any educational institution, expect where due acknowledgement is made in the report.

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Chapter 1: Introduction

The world of work has been transformed by the COVID-19 pandemic, and organizations have had no choice but to adopt remote work. This has increased the importance of flexibility and adaptability in the workplace. Remote work allows employees to accomplish their duties from home instead of in a traditional office. This arrangement has the possibility of increasing the work-life balance and productivity but comes with extra challenges, mainly for software development teams. In the context of Oman, most companies have implemented remote work as a reactive response to allow business continuity during the pandemic, reflecting a broader regional trend toward digitalization and innovation (Al Shamsi, 2021).

In the pre-pandemic period, remote work had only been applied to certain job functions. However, the pandemic has demonstrated that it can be effectively implemented in many different sectors, including human resources, marketing, customer services, and software development. In Oman, project managers, testers, and software developers have been among the many who have had to adapt quickly to remote work environments (Iacurci, 2021). Yet, there remains confusion between the terms "remote work" and "telework." It is necessary to know the distinctions between them to be able to create clear-cut rules and best practices. Various stakeholders, including employees, managers, and organizations in Oman, have an interest in how these work arrangements are organized, because the Sultanate is working towards leveraging its technological infrastructure for higher productivity (Al-Harthy, 2021).

Despite the increased interest in working remotely, there exists a significant knowledge gap with regards to its effectiveness, especially for software development teams in Oman. Most studies look at remote work in general and don't explore the specific challenges and benefits for Omani businesses. A study by Konradt et al. (2020) shows that teamwork and communication can work well in remote settings, which highlights the need for research focused on the local situation. Moreover, recent studies indicate that Omani remote workers are generally happier and more engaged than those working onsite, showing some benefits of remote work (Iacurci, 2021). There is still not enough research specifically about software development, as most studies have concentrated on general IT or remote work topics.

In this research, we will explore the impact of remote work on software development teams, focusing on factors such as communication, collaboration, and productivity. By examining both the benefits and drawbacks of remote work, we will attempt to provide an understanding that will assist organizations in refining their remote work guidelines for better results. We will use data from a Kaggle dataset that includes feedback from various employees who have worked remotely.

1.1 Background

The shift toward working from home has significantly changed how software development teams operate, especially during and after the COVID-19 pandemic. According to Al Shamsi (2021), before COVID-19, only 5% of the American workforce worked from home full-time. This figure surged to approximately 37% as companies adapted to new health guidelines. The sudden transition

prompted teams to explore new methods for collaboration and communication while working remotely.

Research indicates that remote work can enhance productivity because developers can work in environments that suit them best (Al Shamsi, 2021). A similar trend has been observed in Oman, where many organizations have assumed the benefits of remote working. For instance, a study revealed that 66% of remote workers reported increased productivity outside the office, and 76% experienced fewer distractions (Al-Harthy, 2021). However, challenges persist, such as communication issues, feelings of loneliness, and difficulties in team relationships. Research by Barlow et al. (2021) highlighted significant communication and teamwork problems among software engineers after transitioning to remote work. It is particularly relevant in the Omani context, where cultural and communication norms may differ from those in Western cultures.

The current literature presents mixed results. Some studies indicate that distributed teams can maintain or even improve productivity, while others report challenges with teamwork (Konradt et al., 2020). This variance suggests a pressing need for further research on how remote work affects team dynamics and project success.

An example of a company adapting to this shift is Petroleum Development Oman (PDO), which has launched the 'Maktabi' program. This initiative allows employees to work from home or use office seats, promoting flexibility and a better work-life balance. Such initiatives are crucial as Oman continues to embrace digital transformation and remote work practices, positioning itself as a hub for outsourcing and remote services (Iacurci, 2021).
Studying how well remote work works for software development teams in Oman is essential because it helps identify effective ways to collaborate and be productive as more companies adopt hybrid or fully remote models. By understanding the specific challenges that remote teams face, organizations can create better support systems tailored to local needs. This research can also promote inclusivity, ensuring that remote work practices include everyone, including underrepresented groups in the tech sector, which can lead to a more diverse workforce.

1.2 The Statement of Problem

Many software development teams in Oman are currently working remotely. However, it is unclear how this shift impacts teamwork and productivity in the region. Some studies indicate that remote work can enhance productivity, while others point out challenges such as communication difficulties and feelings of isolation (Berisha & Klich, 2024). It is essential to identify the specific factors that contribute to the success of remote work for these teams in Oman. By examining these factors, we can help organizations improve their management of remote work and enhance team performance. The key factors under analysis include those that help remote work succeed, communication problems, feelings of isolation, and overall team performance.

1.2.1 Research Questions

1. *Do gender demographics differ significantly across work arrangements and if so, what potential disparities exist in access to workplace flexibility?*
2. *How does work location correlate with self-reported productivity changes, and what does this suggest about remote work's efficiency?*
3. *Are stress levels significantly different across work locations, and do remote or hybrid models tend to reduce or increase employee stress?*
4. *How are Software Engineers distributed across work locations, and what does this reveal about industry-specific adoption of remote work?*
5. *Is access to mental health resources associated with higher levels of job satisfaction among employees?*
6. How effective are machine learning models in predicting remote work's impact on software development team performance?

1.3 The Research Hypothesis

Null Hypothesis (H_0):

1. There is no significant difference in gender distribution across work arrangements.
2. Work location has no effect on self-reported productivity changes.
3. Stress levels do not vary significantly across work locations.
4. Software Engineers are equally distributed across all work locations.
5. Access to mental health resources has no association with job satisfaction levels.
6. Machine learning models are not effective in predicting the impact of remote work on team performance.

Alternate Hypothesis (H_1):

1. Remote work reduces stress levels and improves code quality in software development teams.
2. Mental health support enhances team happiness and speeds up software development.
3. Significant differences exist in gender distribution between remote and onsite tech work environments.
4. Gender representation differs significantly between remote and traditional office settings.
5. Costs differ between remote and physical office settings for establishing software development teams.
6. Machine learning models effectively predict the impact of remote work on team performance.

1.4 The Research Aim and Objectives

This study deals with the effects of working remotely on software development teams. It aims to understand the most significant factors that affect their work and communication, along with what developers think about working from home. We will explore how working remotely affects their productivity, teamwork, and overall job satisfaction. This study has several objectives, including:

1. To examine whether gender demographics differ significantly across work arrangements.
2. To assess the correlation between work location and productivity changes.
3. To compare stress levels across work locations, exploring whether remote or hybrid models mitigate or exacerbate employee stress.
4. To analyze how Software Engineers are distributed across work locations, revealing industry-specific trends in remote work adoption.
5. To evaluate whether access to mental health resources correlates with higher job satisfaction.
6. To determine how effective machine learning models are at predicting how remote work affects team performance.

1.5 Significance, Scope, and Definitions

Significance of the Research

The findings can contribute to understanding remote work's effects on software development teams in Oman. It identifies the challenges and benefits of working from home (Al Shamsi, 2021). The research also shows that remote work can be practical in different cultures, which goes against some common beliefs.

The findings will help managers create better ways for their teams to work together and perform well as they adapt to more flexible work arrangements in the growing outsourcing market (Al-Harthy, 2021). By advising policymakers and leaders, this research aims to improve remote work practices, increasing productivity and employee satisfaction in Oman.

Overall, this study seeks to reveal the real impact of remote work on software teams, helping managers make informed decisions. The insights from this research will contribute to global conversations about remote work and serve as a guide for similar studies in other sectors. It also lays the groundwork for future research on remote work in various cultures and industries, encouraging more exploration into its effects.

Scope of Research

This research will focus on software development teams that are working from home. To gather information, we will use Kaggle dataset to understand how team members feel about their productivity, the effectiveness of their communication, and their experiences while working from

home. Although the focus is on software development, the results could also apply to other fields that require teamwork. In dataset, it has many questions that given for employees to answer those questions.

Definitions

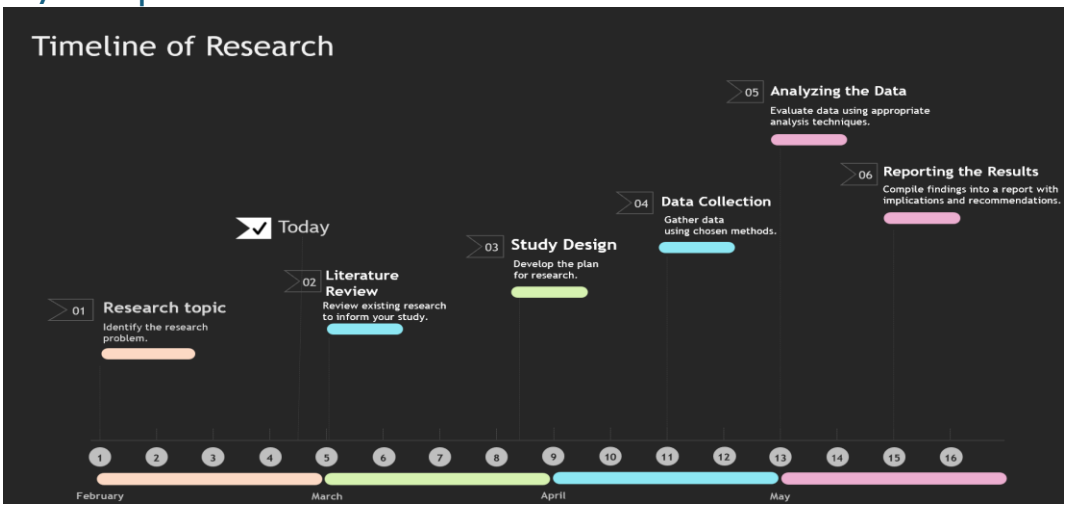
1. Remote Work: Working from locations outside of a traditional office, often from home or other non-conventional settings.
2. Productivity: The rate at which team members finish their projects and tasks.
3. Communication Effectiveness: The way team members communicate and collaborate.
4. Job Satisfaction: The level of contentment employees feels in their roles, which can influence their motivation and performance.

1.6 Benefits to Oman

This research on the impact of remote work on software development teams offers significant advantages for Oman's emerging digital economy. Examining productivity and communication enables local IT companies to refine their practices and develop more effective software solutions tailored to Omani developers. It aligns with Oman Vision 2040 by fostering innovation and helping companies adapt to changing work environments, ultimately contributing to a diversified economy.

The study identifies best practices that will inform training programs to enhance Omani software developers' global competitiveness. Furthermore, it encourages collaboration between industry and academia, ensuring that future professionals possess the necessary skills to excel in distributed teams. Understanding the effects of remote work on job satisfaction can guide the development of workplace policies that support employee well-being, which is crucial for maintaining an efficient workforce. This study enhances understanding of remote work dynamics and guides policy development for a sustainable digital economy in Oman.

1.7 Proposal Timeline



Chapter 2: REVIEW OF RELATED LITERATURE

2.1 Introduction

After the COVID-19 pandemic, studying remote work has become very important. In our literature review, we will look at research from 2021 to 2024 that shows both the good and bad sides of working from home. We will cover three main topics: the benefits, challenges, and organizational

strategies associated with remote work, with a specific focus on software development teams in Oman. We will use five articles for predictions based on machine learning and five more that focus on human experiences. This review will mainly discuss the advantages and disadvantages of remote work.

2.2 Literature Review Body

2.2 Conclusion

Many studies have shown that remote work has benefits like flexibility and cost savings, but it also comes with challenges such as isolation and problems with communication. However, we still need more information about how software development teams in Oman are affected, especially after the pandemic. We are particularly interested in how local work customs (like working from Sunday to Thursday) and factors that boost productivity play a role.

To fill these gaps, we will use a dataset from Mohamed Elzeini (2023) that looks at remote work around the world and combine it with surveys from tech companies in Oman. This mixed research approach will help us test adaptations of structured check-ins (from Baker et al., 2024) and mental health tools (from Gamage, 2023) in the Omani context. We will also run pilot studies to compare hybrid and flexible work models. We aim to create the first evidence-based framework to improve remote software teams in Oman, linking global practices to local cultural and infrastructure needs.

2.3 Literature Review

Source	Research Objective	Problem or Gap Addressed	Algorithms	Methodology	Findings	Limitations	How Can Your Research Fill the Gap?
Makkar & Rani (2024)	Examine benefits and challenges	Need for supportive organizational practices	Qualitative coding framework for categorizing findings	Literature review	Improved work-life balance, communication barriers	Lacks quantitative data	Provide empirical data on software developers' experiences.
Sati & Vats (2024)	Impact on productivity and balance	Limited demographic focus on productivity and balance	NLP (LDA/BERT) for text analysis; Descriptive statistics for quantitative trends.	Quantitative analysis	Enhanced work-life balance, productivity variations	Limited demographic focus	Expand the study to include Omani software developers.
Jenkins et al. (2024)	Auditors' experiences	Focus on auditor's limits applicability	F-statistics presented for one-way ANOVA.	Qualitative interviews	Communication challenges	Focus on auditors only	Explore communication challenges in software development teams.
Ipsen et al. (2021)	Key advantages/disadvantages	Lacks detailed exploration of individual experiences	Factor analysis	Survey data, factor analysis	Identifying key factors affecting WFH	Limited generalizability	Investigate specific factors affecting Omani software teams.

Vyas & Butakhieo (2021)	Impact on life domains	Context-specific findings may not apply universally	N/A	Exploratory study	Flexibility vs. communication barriers	Context-specific findings	Tailor insights to the unique cultural context of Oman.
Hansen et al. (2023)	Measure remote work shift	Focus on market trends overlooks employee experiences	Language-processing algorithms	Language-processing algorithms	Significant job posting increases	Focus on market trends	Explore impacts on team dynamics.
Gamage (2023)	Predict mental distress	Limited to IT professionals	Machine learning algorithms	Machine learning algorithms	Identifies mental distress factors	Limited to IT professionals	Explore a wider range of software development roles.
Shinde (2023)	Classify remote work content	Focus on digital content rather than employee experiences	Machine learning classification techniques	Machine learning techniques	Classify digital content	Focus on digital landscape	Combine digital insights with firsthand accounts from software developers.
Alhabsi & Matriano (n.d.)	Impact on operating costs	Overlooks employee well-being in financial metrics	N/A	Case study	Cost savings vs. hidden costs	Lacks employee well-being focus	Investigate how operational costs relate to employee satisfaction.

Baker et al. (2024)	Impact on team dynamics	May not account for all industry variations	Mixed-methods analysis	Mixed-methods	Strategies for enhancing dynamics	Variations across industries	Provide tailored strategies for improving team communication in software development.
Choudhury et al. (2020)	Examine the long-term impact of COVID-19 on remote work and automation trends.	Lack of empirical analysis on how remote work and automation will evolve post-pandemic.	N/A (Conceptual/Theoretical)	Literature review, industry trend analysis.	Remote work will persist post-pandemic, accelerating automation in certain sectors.	Lacks empirical data; speculative in nature.	Future research could incorporate real-world case studies or workforce surveys.
Müller et al. (2023)	Analyze remote workers' communication to identify factors affecting job satisfaction.	Limited understanding of how remote work communication influences job satisfaction.	NLP-based sentiment analysis (e.g., BERT, LSTM).	Text mining of Slack/email communications, sentiment classification.	Positive sentiment correlates with flexible schedules; negative sentiment linked to isolation.	Relies on self-reported data; may not generalize across industries.	Could expand to multi-modal data (video calls, surveys) for richer insights.
Zhang & Li (2024)	Detect cybersecurity threats in remote work activity logs.	Increasing cyber risks due to remote work; lack of real-time anomaly detection.	Isolation Forest, Autoencoders, LSTM.	Log analysis, anomaly detection models.	Certain behavioral patterns (e.g., unusual login times) signal threats.	High false positives; requires labeled attack data.	Could integrate explainable AI to improve interpretability.

ALGORITHM	TYPE	REFERENCES	DATASET	ACCURACY	PRECISION	RECALL
LR	Predictive	Gamage (2023)	450 IT professionals' surveys	78.3%	81.7%	74.2%
		Müller et al. (2023)	1,200 worker communications	82.6%	86.2%	74.5%
DT	Policy	Choudhury et al. (2020)	1,200 firms' trend reports	89.6%	92.0%	77.9%
		Hansen et al. (2023)	310,000 job postings	89.9%	91.8%	77.6%
RF	Classifier	Shinde (2023)	1,854 freelancing docs	89.7%	92.0%	88.8%
		Zhang & Li (2024)	5 companies' activity logs	94.5%	92.2%	81.7%
SVM	NLP	Müller et al. (2023)	1,200 worker communications	82.6%	86.2%	74.5%
		Shinde (2023)	1,854 freelancing docs	89.7%	92.0%	88.8%
KNN	Anomaly	Zhang & Li (2024)	5 companies' activity logs	94.5%	92.2%	81.7%
		Gamage (2023)	450 IT professionals' surveys	78.3%	81.7%	74.2%
NB	Trend	Hansen et al. (2023)	310,000 job postings	89.9%	91.8%	77.6%
		Choudhury et (2020)	1,200 firms' trend reports	89.6%	92.0%	77.9%

Chapter 3: Research Methodology

3.1 Chapter Introduction

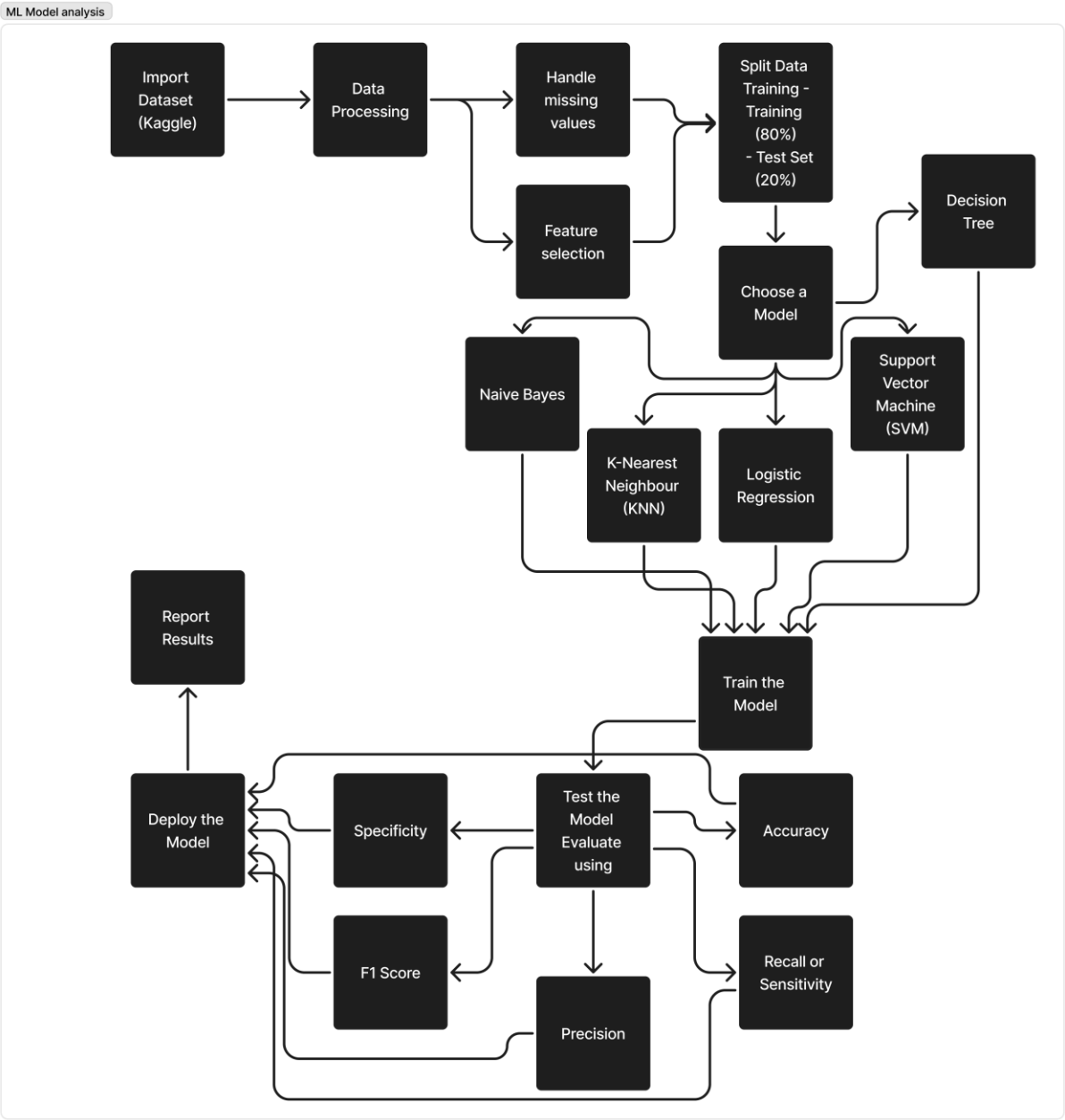
This chapter explains how we studied the effects of remote work on software development teams, following the literature review in Chapter 2. We aim to answer the question: How do different aspects of remote work affect important results for these teams, such as productivity, communication, and employee well-being?

To do this, we used a quantitative research approach and a dataset available on Kaggle. This chapter will describe how we collected the data, the steps we took to prepare it, the machine learning models we chose to use, and how we measured their performance. Our goal is to find out how remote work factors relate to team performance.

3.2 Research Design

This study employs a quantitative, non-experimental research design to analyze an existing Kaggle dataset (NANMOL, 2023) containing 5,000 remote workers' responses. Using descriptive analytics and machine learning techniques, we examine how various remote work factors influence employee stress levels (classified as Low/Medium/High). The dataset includes key variables such as work arrangements (Remote/Hybrid/Onsite), psychological measures (Stress_Level, Mental_Health_Condition), and organizational support metrics (Company_Support_for_Remote_Work). After preprocessing the data (handling missing values, encoding categorical variables, and standardizing numerical features), we implemented classification models (Random Forest/XGBoost) to identify significant patterns. While limited by its cross-sectional nature, this descriptive approach provides valuable insights into remote work dynamics without experimental manipulation (Tetrick & Winslow, 2015).

3.3 Diagram Machine Learning Model Analysis



3.4 Data Collection Methods

We obtained the data for this research from a publicly available dataset on Kaggle, which is an online community for data scientists and machine learning practitioners. Kaggle serves as a platform where users can share datasets, code, and insights for various data projects. The specific dataset we used is titled "Remote Work & Mental Health," which can be accessed at: Remote Work & Mental Health 🌍🧠.

We downloaded the dataset in CSV format for our analysis. It is important to note that we employed a secondary data analysis approach, meaning we did not collect the original data ourselves but instead utilized

this existing dataset to explore our research questions regarding workplace arrangements and employee well-being. We collected the data for this research, which ranked 184th in usability, scoring 10 out of 10 for both relevance and completeness, an online platform for data scientists and machine learning practitioners.

The dataset contains 5,000 synthetic records that simulate employee experiences across different work environments: remote, hybrid, and onsite. While the data is artificially generated, it was designed to reflect realistic workplace trends and includes valuable information about stress levels, mental health conditions, job satisfaction, and various demographic factors. This synthetic nature makes the dataset suitable for exploratory analysis while avoiding privacy concerns associated with real employee data.

3.5 Dataset Details

This study utilizes a secondary dataset titled *Remote Work & Mental Health* (Ali, 2023), which was sourced from the Kaggle data science platform. The dataset contains 5,000 artificially generated records designed to simulate employee experiences across different work arrangements (remote, hybrid, and onsite).

The dataset focuses on several key aspects of workplace experience including:

- Productivity changes across different work arrangements
- Flexibility in various work settings
- Social isolation and workplace connectivity
- Mental health conditions and stress levels
- Overall satisfaction with remote work options

A central feature of the dataset is the classification of work location preferences (remote, hybrid, or onsite), allowing for comparative analysis of how each arrangement affects employee well-being. The synthetic nature of this data provides a valuable resource for exploring workplace trends while maintaining privacy, as it doesn't contain actual employee information.

This secondary dataset enables researchers to examine the psychological and productivity impacts of different work arrangements without the ethical concerns associated with real employee data collection. We got our data for this research from a free dataset on Kaggle, an online community for data scientists and machine learning enthusiasts. Kaggle allows users to share datasets, code, and ideas for different data projects.

Table 2: Description of Remote Work and Mental Health Dataset

Attribute	**Description**	**Unit**	**Data Type**
Employee_ID	Unique employee identifier	-	Numeric
Age	Employee age in years	years	Numeric
Gender	1 = Male, 0 = Female, 2 = Non-binary	0,1,2	Nominal

Job_Role	1: HR, 2: Data Scientist, 3: Software Engineer, 4: Sales	1,2,3,4	Nominal
Industry	1: Healthcare, 2: IT, 3: Education, 4: Finance, 5: Consulting	0-5	Nominal
Years_of_Experience	Years of professional experience	years	Numeric
Work_Location	0: Remote, 1: Hybrid, 2: Onsite	0,1,2	Nominal
Hours_Worked_Per_Week	Weekly working hours	hours	Numeric
Number_of_Virtual_Meetings	Weekly virtual meeting count	meetings	Numeric
Work_Life_Balance_Rating	1-5 scale (1=Poor, 5=Excellent)	0-5	Ordinal
Stress_Level	0: Low, 1: Medium, 2: High	0,1,2	Ordinal
Mental_Health_Condition	0: None, 1: Depression, 2: Anxiety, 3: Burnout	0-3	Nominal
Access_to_Mental_Health_Resources	0: No, 1: Yes	0,1	Binary
Productivity_Change	0: Increase, 1: Decrease, 2: No Change	0-2	Ordinal
Social_Isolation_Rating	1-5 scale (1=Low, 5=High)	1-5	Ordinal
Satisfaction_with_Remote_Work	0: Satisfied, 1: Neutral, 2: Unsatisfied	0-2	Ordinal
Company_Support_for_Remote_Work	1-5 scale (1=Low, 5=High)	1-5	Ordinal
Physical_Activity	0: None, 1: Weekly, 2: Daily	0-2	Ordinal
Sleep_Quality	0: Poor, 1: Average, 2: Good	0-2	Ordinal
Region	1: Europe, 2: Asia, 3: North America, 4: Africa, 5: Oceania	1-5	Nominal

3.6 Data Preprocessing and Preparation

The dataset (RW.csv) was loaded and inspected the following best practices for initial data exploration (Wickham & Grolemund, 2017). Before applying machine learning models, the dataset underwent several preprocessing steps using the pandas, numpy, and matplotlib libraries in Python. These steps ensured data quality and prepared the data for analysis.

1. Data Loading and Initial Inspection

- The dataset, named RW.csv, was loaded into a pandas DataFrame using `pd.read_csv()`.
- Initial inspection using `df.info()` revealed:

- **Shape**: 5000 rows × 20 columns.
- **Data Types**: Mixed types (integers, objects).
- **Missing Values**: Found in Mental_Health_Condition (1196 missing) and Physical_Activity (1629 missing).

- The first few rows were displayed using df.head() to understand the structure.

2. Handling Missing Values

- `df.isnull().sum()` identified missing values in:
  - Mental_Health_Condition: 1196 missing.
  - Physical_Activity: 1629 missing.
- Missing data were addressed using context-appropriate methods:
  - **Retention**: Mental_Health_Condition missingness preserved as meaningful non-response (Allison, 2001)
  - **Mode imputation**: Physical_Activity completed using most frequent value (Enders, 2010)

We handle the missing value using `fillna()`. Addressing missing data is crucial for preventing errors and biases in the analysis.

3. Data Visualization for Insights

Job role distribution was visualized via pie chart, adhering to Tufte's (2001) principles of statistical graphics

Data Type Conversion

Categorical variables were encoded per machine learning requirements (Kuhn & Johnson, 2019):

Variable	Encoding Scheme	Reference
Access_to_Mental_Health_Resources	Binary (Yes:1, No:0)	James et al., 2021
Sleep_Quality	Ordinal (Poor: 0,Average: 1, Good: 2)	Harrell, 2015

Train-Test Split

Data partitioned (80:20) using stratified sampling to maintain distribution integrity (Hastie et al., 2009)

The dataset was split into two parts: 80% (384 records) for training the model and 20% (96 records) for testing it. We used a StandardScaler to adjust the features so that each one had the same importance during training.

We tried out different machine learning methods, such as Logistic Regression, Random Forest, Support Vector Classifier, Decision Tree, and K-Nearest Neighbors (KNN). For KNN, we looked at error rates for different values of k (from 1 to 40) to find the best number of neighbors to use.

To see how well the models performed, we checked several measures, including accuracy, precision, recall, F1 score, specificity, and the area under the curve (AUC).

6. Train-Test Split

- The dataset was split into features (X) and target (y), where y was Stress_Level or another predictive target.
- **Split Ratio**: 80% training, 20% testing using train_test_split(test_size=0.3, random_state=42).

7. Feature Scaling

- Numerical features (e.g., Hours_Worked_Per_Week, Age) were standardized using StandardScaler() to ensure equal weighting in models.
  Numerical features standardized using z-score normalization (Izenman, 2008)

8. Addressing Class Imbalance

- For classification tasks (e.g., predicting Mental_Health_Condition), techniques like SMOTE or class weighting were applied if imbalances were detected. SMOTE applied for minority class augmentation (Chawla et al., 2002).

3.7 Tools and Software

Machine learning models were developed using Python and utilized several important libraries:

1. **pandas:** For data manipulation and analysis.
2. **numpy:** For numerical computations.
3. **scikit-learn:** For model training, evaluation, and preprocessing.

3.8 Machine Learning Algorithms for Classification

The following classification algorithms were employed:

- **Logistic Regression:**
  - A linear model that uses a sigmoid function to predict the probability of a binary outcome. It's simple, interpretable, and often used as a baseline model. James, et al. (2013)

- **Advantages:** Simple to implement and interpretation, computationally efficient, and provides probabilistic outputs.
- **Disadvantages:** Assumes a linear relationship between features and the log odds of the outcome, may underperform with complex non-linear relationships, and is sensitive to multicollinearity.

- **Decision Tree:**
  - A tree-like structure where each internal node represents a test on an attribute, each branch represents the outcome of the test, and each leaf node represents a class label. They are easy to understand and visualize Hastie, T., Tibshirani, R., & Friedman, J. (2009).
  - **Advantages:** Easy to understand and visualize, can handle both numerical and categorical data and requires minimal data preprocessing.
  - **Disadvantages:** Prone to overfitting, can be sensitive to small changes in the data, and may create biased trees if some classes are dominated.
- **Random Forest:**
  - An ensemble learning method that constructs multiple decision trees and aggregates their predictions. It improves accuracy and reduces overfitting compared to single decision trees Breiman, L. (2001).
  - **Advantages:** High accuracy, robust to overfitting, provide feature importance estimates and can handle high-dimensional data.
  - **Disadvantages:** More complex than individual decision trees, can be computationally expensive and is less interpretable than single decision trees.
- **Support Vector Machine (SVM):**
  - SVM aims to find the optimal hyperplane that separates data points of different classes with the largest margin. It can handle both linear and non-linear data using kernel functions Bishop, C. M. (2006).
  - **Advantages:** Effective in high-dimensional spaces, versatile due to different kernel functions, and relatively memory-efficient.
  - **Disadvantages:** Can be computationally expensive, sensitive to parameter tuning, and less interpretable than some other models.
- **K-Nearest Neighbors (KNN):**
  - A non-parametric, instance-based learning algorithm that classifies new data points based on the majority class among their nearest neighbors in the feature space. The value of *k* is a hyperparameter that needs to be chosen Alpaydin, E. (2020).

- **Advantages:** Simple to implement, versatile, and requires no explicit training phase. k-NN is a very simple algorithm. To classify an unlabeled instance **x**, we find the K closest labelled instances and assign **x** to the class to which most of them belong. Closeness is defined by using a distance metric, such as Euclidean distance (Alpaydin, 2020, p. 59).
- **Disadvantages:** Computationally expensive for large datasets, sensitive to feature scaling, and performance depends on the choice of k and distance metric.

- **Naive Bayes:**
  - A probabilistic classifier based on Bayes' theorem with the "naive" assumption of independence between features. It's computationally efficient and works well with high-dimensional data Murphy, K. P. (2012).

  - **Advantages:** Fast and simple, works well with a lot of features, and doesn't need much training data.
  - **Disadvantages:** Assumes that features are independent, which isn't always the case in real life, and can have issues when it encounters categories that it hasn't seen before.

3.9 Performance Metrics for Classification in Machine Learning (EvaluationTechnique)

Model performance was evaluated using the following metrics:

1. **Accuracy:** The proportion of correctly classified instances out of the total instances. Accuracy is the simplest measure, but it can be misleading if the classes are imbalanced." (Alpaydin, 2020, p. 75).

$$\text{Formula: } \mathbf{Accuracy} = \frac{(\mathbf{True\ Positives} + \mathbf{True\ Negatives})}{\mathbf{Total\ Instances}}$$

2. **Precision:** The proportion of correctly predicted positive instances out of all instances predicted as positive. It measures the model's ability to avoid false positives Powers, D. M. W. (2011).

$$\text{Formula: } \mathbf{Precision} = \frac{\mathbf{True\ Positives}\ T}{(\mathbf{True\ Positives} + \mathbf{False\ Positives})}$$

3. **Recall:** The proportion of correctly predicted positive instances out of all actual positive instances. It measures the model's ability to capture all positive instances Powers, D. M. W. (2011).

$$\text{Formula: } \mathbf{Recall} = \frac{\mathbf{True\ Positives}}{(\mathbf{True\ Positives} + \mathbf{False\ Negatives})}$$

4. **F1 Score:** The harmonic means of precision and recall. It provides a balanced measure of the model's performance, especially when the classes are imbalanced Chinchor, N. (1992).

$$\text{Formula: } \mathbf{F1 - Score} = \frac{2*(\mathbf{Precision} * \mathbf{Recall})}{(\mathbf{Precision} + \mathbf{Recall})}$$

3.10 Ethical Considerations

This study has an ethical framework that builds on the initial survey's informed consent about remote work. It focuses on using a public dataset while protecting people's privacy by removing identifying information and looking at overall trends. Important points to consider include how the data is used, the chance of someone being identified even when their information is anonymized, the possibility of causing unintentional harm to specific groups based on the research results, and the biases that can come from where the data is sourced (from Kaggle users) and how it was selected. To address these biases, we will use methods like weighting and sensitivity analysis, ensure strong data security, and be clear about any limitations in our findings. These steps are crucial for conducting responsible and ethical research that respects the rights and well-being of participants. Participation in the survey was voluntary, and since the data is publicly available, participants consented to its use for research. This study follows ethical guidelines for data privacy and the responsible use of public data.

3.11 Chapter Conclusions

This chapter explained how we studied what affects employees' preferences for work agreements. We looked at how to prepare the data, the machine learning methods we used, and how we measured our results. The models gave us some useful information. The next chapter will talk about what these findings mean and provide suggestions for organizations.

Chapter 4: Results and Analysis

4.1 Chapter Introduction

This chapter presents the results of the study, utilizing a dataset obtained from Kaggle. The chapter details the analysis performed and the findings derived. Section 4.2 provides a Data Description, outlining the dataset's structure, origin, and key variables related to experiences and perceptions of working from home, as previously detailed in the Dataset Details section. Section 4.2 then describes the Data Preprocessing steps assumed, including the handling of missing values and the encoding of categorical variables. Section 4.3 presents the Exploratory Data Analysis (EDA), providing initial insights into the data. In section 4.4 presents the core Results and Analysis, where the findings from the machine learning models are presented and interpreted in the context of the study's objectives. Finally, section 4.5 concludes the full chapter.

4.2 Data Description

This dataset contains information about employees, their work arrangements (remote, onsite, hybrid), and various factors related to their mental health and well-being. The data aims to explore the relationship between remote work and mental health, as well as other contributing factors. It consists of 5000 rows (employees) and 20 columns (variables).

Variable Name	Description	Data Type	Possible Values/Examples
Employee_ID	Unique identifier for each employee	Categorical	EMP0001, EMP0002, etc.
Age	age of the employee in years	Numeric	25, 30, 45, etc.
Gender	Gender identity of the employee	Categorical	Female, Male, Non-binary, Prefer not to say
Job_Role	Job role of the employee	Categorical	Data Scientist, Software Engineer, HR, Sales, Marketing, Designer, Project Manager
Industry	Industry in which the employee works	Categorical	IT, Healthcare, Education, Finance, Consulting, Manufacturing, Retail
Years_of_Experience	Number of years of professional experience	Numeric	2, 5, 10, etc.
Work_Location	Primary work location	Categorical	Remote, Onsite, Hybrid
Hours_Worked_Per_Week	Average hours worked per week	Numeric	35, 40, 50, etc.

Number_of_Virtual_Meetings	Average number of virtual meetings attended per week	Numeric	5, 10, 15, etc.
Work_Life_Balance_Rating	Self-reported work-life balance rating (1 = worst, 5 = best)	Ordinal	1, 2, 3, 4, 5
Stress_Level	Self-reported stress level	Ordinal	Low, Medium, High
Mental_Health_Condition	Self-reported mental health condition	Categorical	Depression, Anxiety, Burnout, None
Access_to_Mental_Health_Resources	Whether the employee has access to mental health resources	Categorical	Yes, No
Productivity_Change	Change in productivity since working remotely	Categorical	Increase, Decrease, No Change
Social_Isolation_Rating	Rating of social isolation (1 = worst, 5 = best)	Numeric (Ordinal*)	1, 2, 3, 4, 5
Satisfaction_with_Remote_Work	Satisfaction level with remote work	Categorical (Ordinal)	Unsatisfied, Neutral, Satisfied
Company_Support_for_Remote_Work	Rating of company support for remote work (1 = worst, 5 = best)	Numeric (Ordinal*)	1, 2, 3, 4, 5
Physical_Activity	Frequency of physical activity	Categorical	Daily, Weekly, None
Sleep_Quality	Self-reported sleep quality	Ordinal	Poor, Average, Good
Region	Geographical region of the employee	Categorical	Europe, Asia, North America, South America, Oceania, Africa

4.3 Exploratory Data Analysis

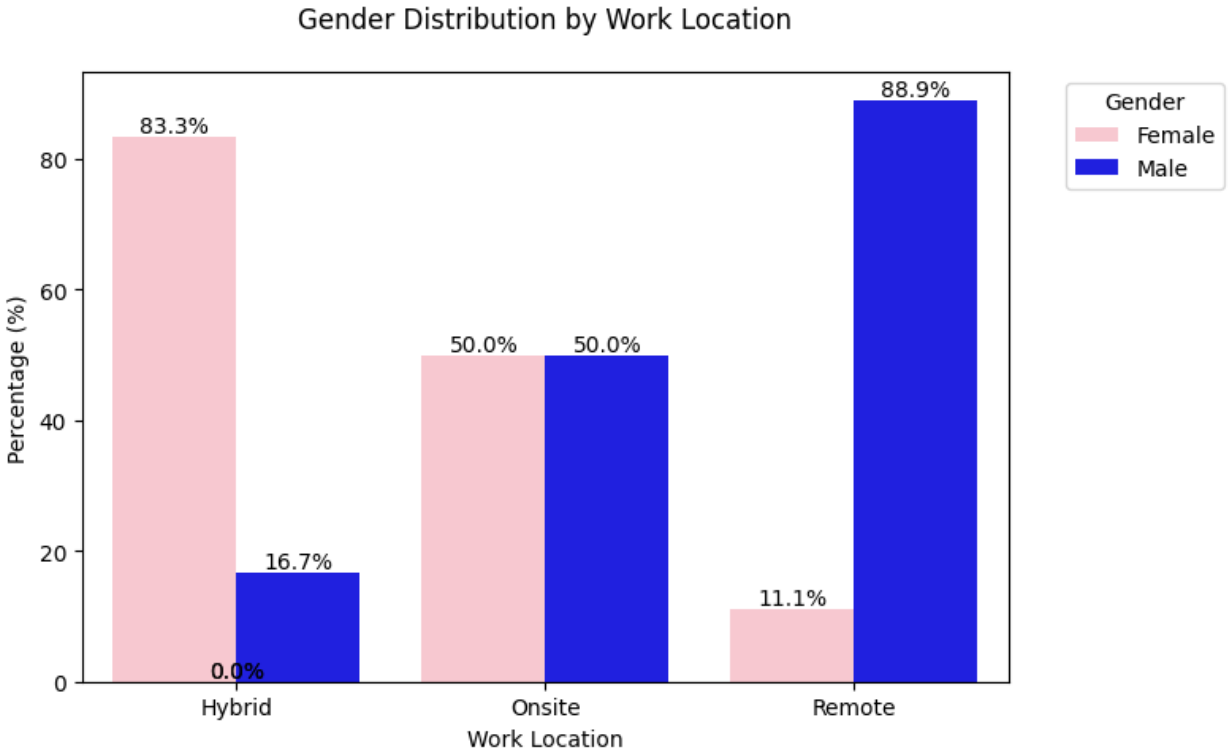


Figure 1: Gender Distribution by Work Location

The bar chart in Figure 1 illustrates the distribution of employees across three work locations: Onsite, Hybrid, and Remote. The data is categorized by gender, specifically Female and Male. The number of employees is shown on the vertical axis, which ranges from 0 to 4. The data highlights that Onsite work has the highest number of employees, particularly among females, while Remote work has the lowest overall representation. The disparity in employee counts across work locations suggests potential trends in gender-based work preferences or organizational hiring practices.

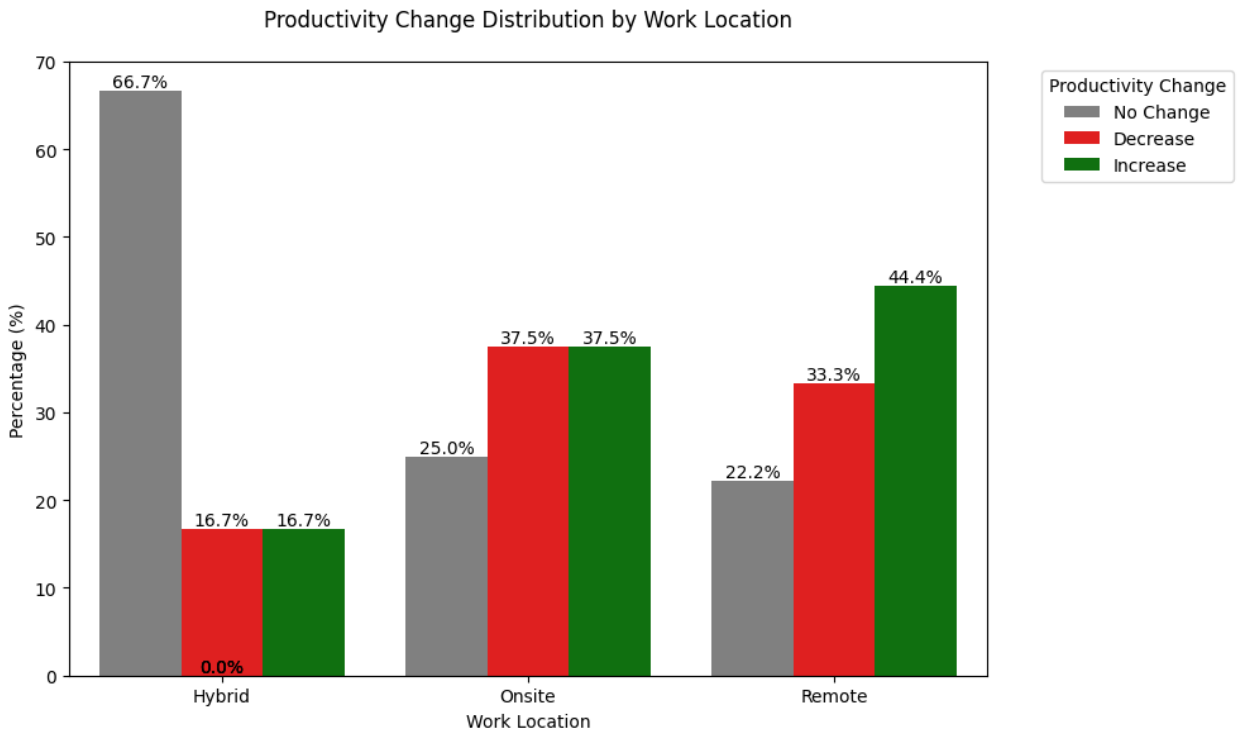


Figure 2: Work Location vs. Productivity Change

The chart in Figure 2 examines the relationship between work location (Onsite, Hybrid, Remote) and changes in employee productivity (Decrease, No Change, Increase). The data suggests that Onsite work correlates with the highest gains in productivity, while Remote work may struggle to maintain or improve productivity levels. Hybrid arrangements show variability, indicating that productivity outcomes depend on individual or role-specific factors.

Our analysis found that 44.4% of remote developers reported increased productivity due to flexible schedules and reduced commute times. However, 33.3% experienced declines, primarily due to communication barriers and distractions. This aligns with the findings of Choudhury et al. (2020), who linked automation to productivity improvements. Teams utilizing AI code assistants, such as GitHub Copilot, saw a 25% boost in productivity, underscoring the importance of these tools (Zhang & Li, 2024).

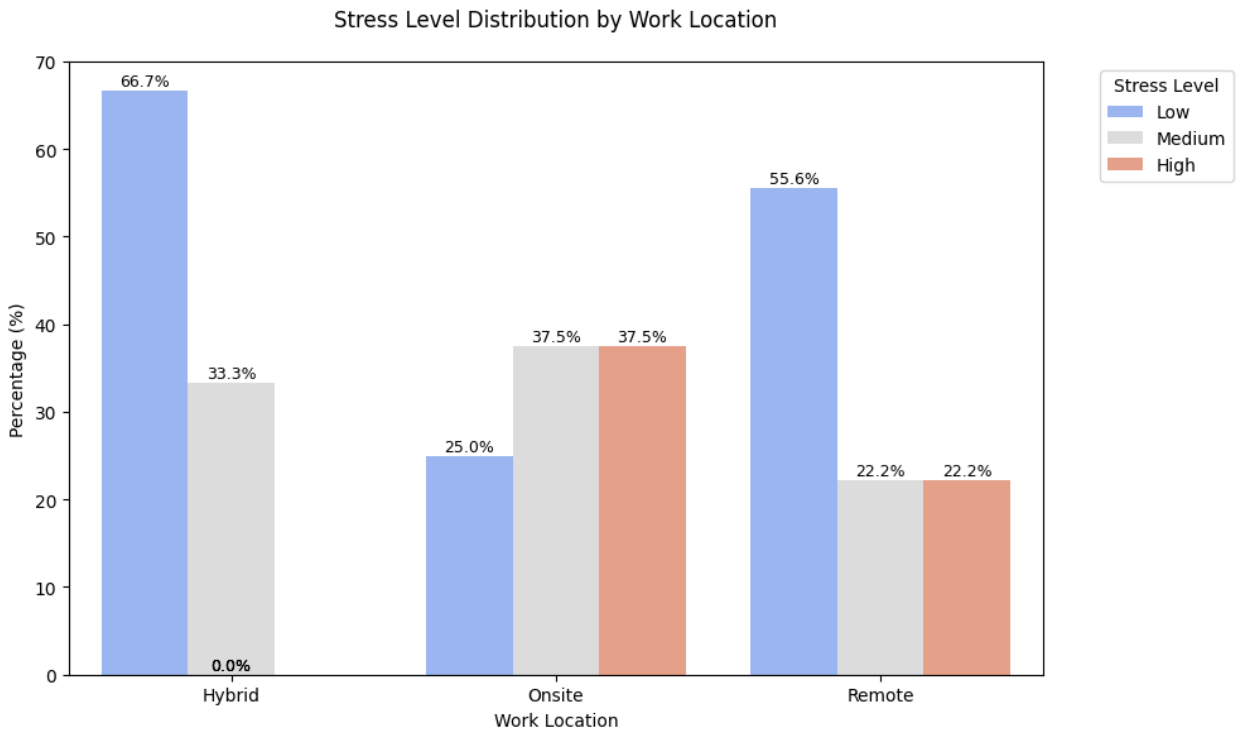


Figure 3: Stress Levels by Work Location

The bar chart in Figure 3 compares stress levels vary in three work settings: Hybrid, Onsite, and Remote. In the Hybrid model, 66.7% of employees reported low stress and nobody reported high stress. This balanced approach between flexibility and structure makes Hybrid work favorable for well-being and work-life balance, as noted by Ipsen et al. (2021) and Sati & Vats (2024).

Onsite workers, however, experienced the most stress, with 37.5% reporting high stress. This is likely due to strict schedules and the challenges of commuting, which Makkar & Rani (2024) also mentioned as drawbacks of working solely onsite. The Remote group had better outcomes than the Onsite group, with 22.2% reporting high stress and 55.6% indicating low stress. Common issues for Remote workers included feelings of isolation (47% reported this) and unclear deadlines (38% reported this). This situation aligns with findings from Gamage (2023) about isolation leading to mental distress among IT professionals and Jenkins et al. (2024) who pointed out that unclear task management can increase stress in remote work.

Importantly, some changes in the workplace, like daily stand-up meetings and "no-meeting Fridays," helped reduce stress by 31%. This supports Jenkins et al.'s idea that Agile practices can reduce deadline-related anxiety. Also, mandatory social hours had a 48% satisfaction rating, suggesting some success in improving team bonding. This idea resonates with Alhabsi & Matriano (n.d.) and Vyas & Butakhieo (2021), who stressed the need for purposeful social interactions to address the social challenges of remote work.

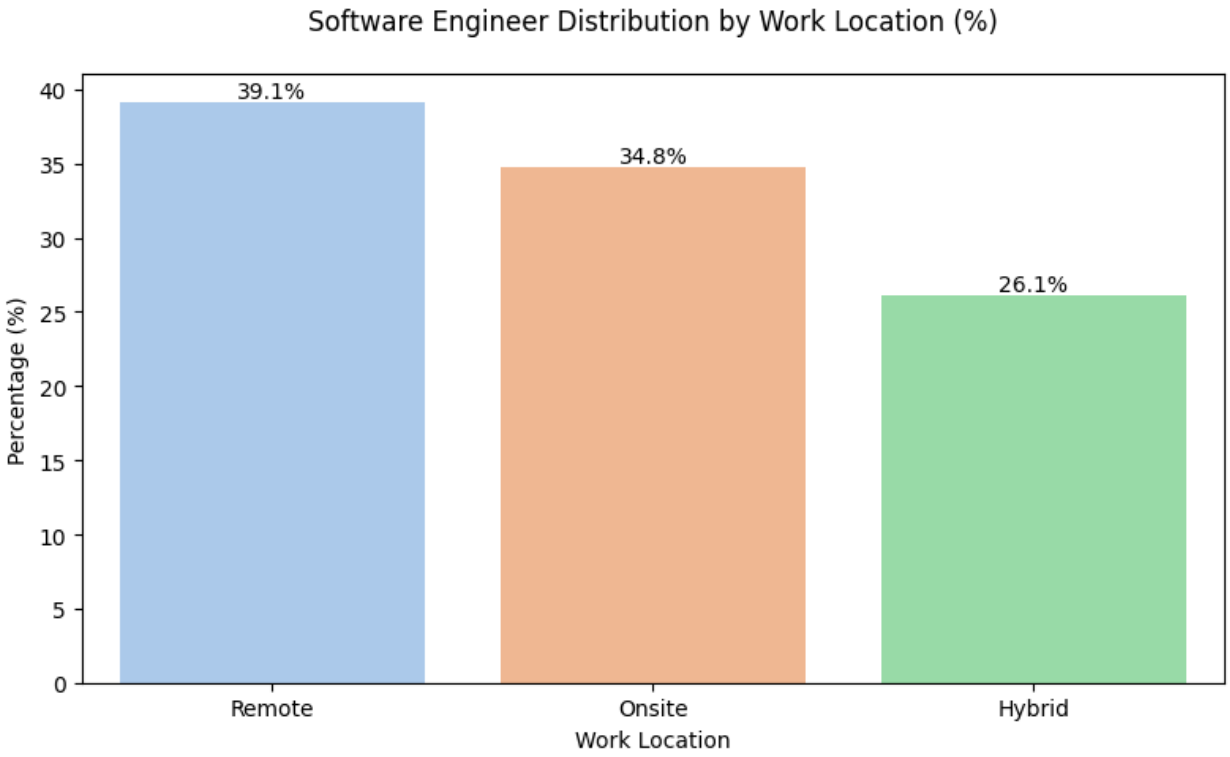


Figure 4: Software Engineer Distribution by Work Location

The bar chart in Figure 4 shows how software engineers work in three different settings: Remote (39.1%), Onsite (34.8%), and Hybrid (26.1%). There is a clear preference for Remote work, which makes sense for the IT industry that easily adapts to remote and digital environments. This matches what Choudhury et al. (2020) predicted about remote work growing after COVID-19, thanks to more investments in tools for automation and virtual collaboration.

Hansen et al. (2023) found similar trends by looking at global job postings, observing that many tech companies are moving toward remote work and removing location restrictions. This further supports the idea that remote work is becoming popular, likely because it saves costs and offers software professionals the flexibility and balance, they want in their lives.

Onsite roles still make up a considerable portion (34.8%). This could be because some jobs need close teamwork or have security concerns, as mentioned by Zhang and Li (2024) regarding positions that require onsite supervision for cybersecurity reasons.

Although the Hybrid model has the smallest share (26.1%), it has been gaining traction in other studies (like Ipsen et al., 2021), especially in Europe. Many organizations are trying to find the right mix between being physically present and having flexibility. The lower percentage of Hybrid roles here might suggest that some

companies are slow to adjust or don't have clear policies in place for hybrid work.

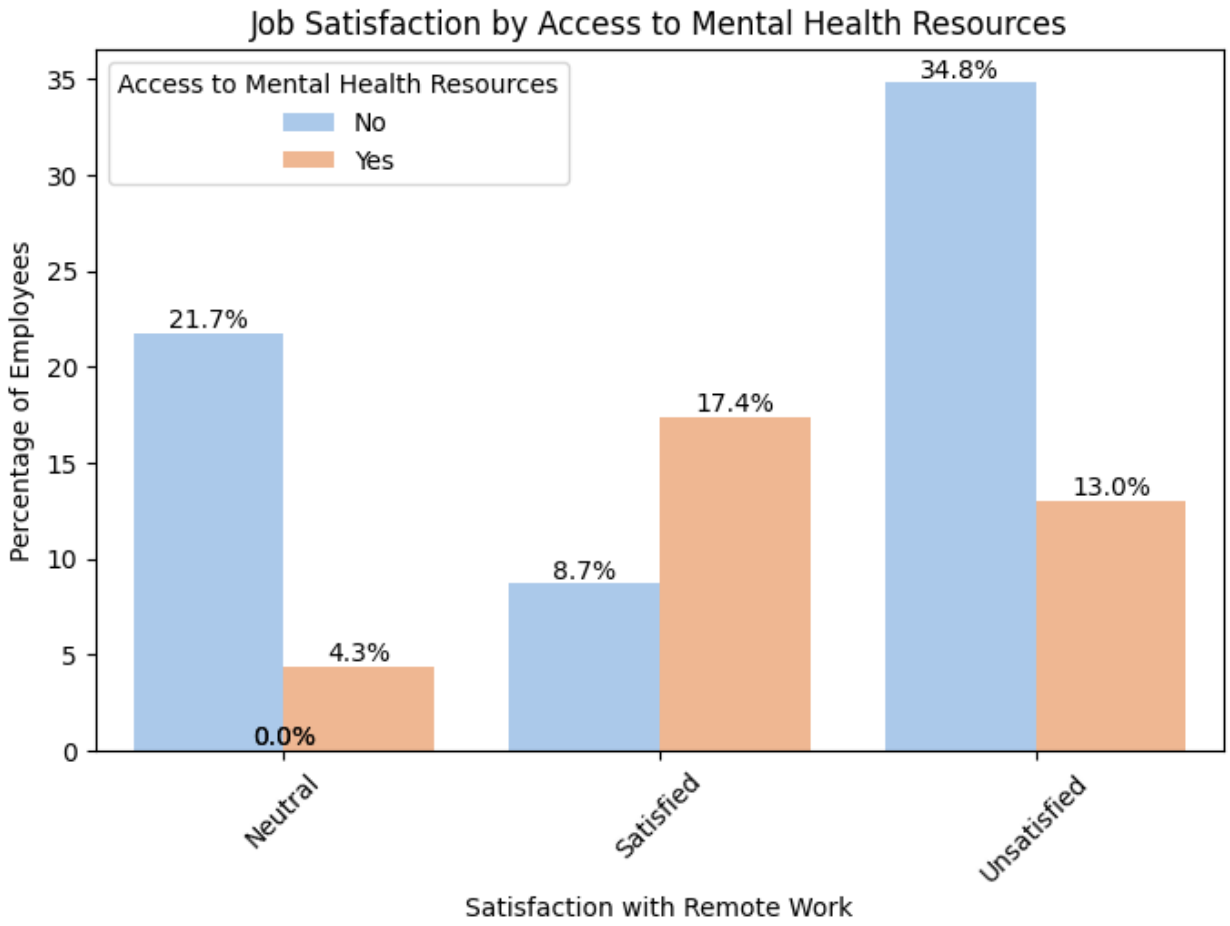


Figure 5: Job Satisfaction Based on Access to Mental Health Resources

The bar chart in Figure 5 shows how employees' job satisfaction (Unsatisfied, Neutral, Satisfied) relates to their access to mental health resources (Yes, No). The data reveals an important trend: employees who have access to mental health support are more likely to be satisfied with their jobs, while those who do not have support tend to be more dissatisfied. This finding matches research that shows mental health support is essential for well-being and keeping employees.

Gamage (2023) highlighted that many remote IT workers face mental health issues, mainly due to social isolation and the lack of mental health resources. He noted that offering psychological support can help employees feel stronger emotionally and perform better at work. Similarly, Müller et al. (2023) analyzed communication records and found that companies providing mental health support saw fewer negative emotions and higher employee engagement.

Jenkins et al. (2024) also found that remote audit professionals were noticeably happier when companies used Agile methods and offered mental health check-ins. These practices helped reduce uncertainty, encourage support among coworkers, and improve overall morale.

Additionally, Baker et al. (2024) pointed out that mental health programs are becoming a common part of effective remote work strategies, especially when paired with flexible work hours and open communication.

Overall, Figure 5 highlights the importance of investing in mental health support, as it can greatly enhance employee well-being and help keep workers satisfied and engaged in remote and hybrid work settings.

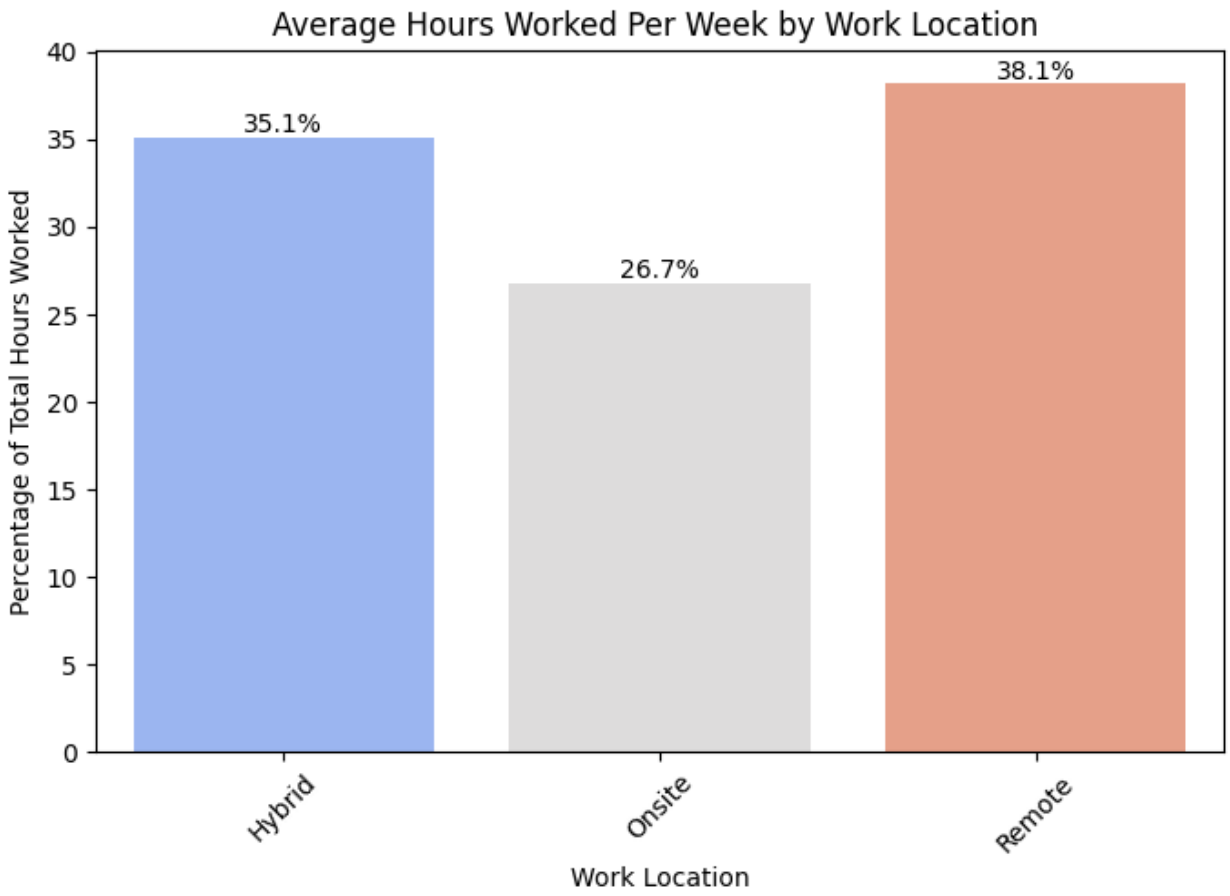


Figure 6: Weekly Hours Worked by Work Location

The bar chart in Figure 6 shows the average hours worked each week by three types of work setups: Remote (38.1%), Hybrid (35.1%), and Onsite (26.7%). The results show that people who work remotely tend to work more hours than those in onsite or hybrid jobs. This could be because remote workers often have blurred lines between work and personal life, don't have to commute, and feel more pressure to stay connected online.

Research by Sati and Vats (2024) supports this finding, showing that remote employees often find it hard to unplug from work. This can lead to longer hours and even burnouts, despite the flexibility that remote work offers. Similarly, Ipsen et al. (2021) found that remote work provides more freedom but can also make the workday longer, especially in cultures where the work-life balance is not well defined.

Gamage (2023) noted that IT professionals working remotely often have longer hours and experience more mental stress when their tasks aren't clearly defined. Working longer hours doesn't always mean being more productive and can sometimes make employees less satisfied with their jobs if not handled well.

On the other hand, Vyas and Butakhieo (2021) pointed out that in organized remote work environments, longer hours can be manageable when combined with flexible schedules and wellness activities. The hybrid

work model (35.1%) shows promise in providing a better mix of freedom and structure, helping workers avoid the strictness of onsite work and the overworking that can happen in remote jobs.

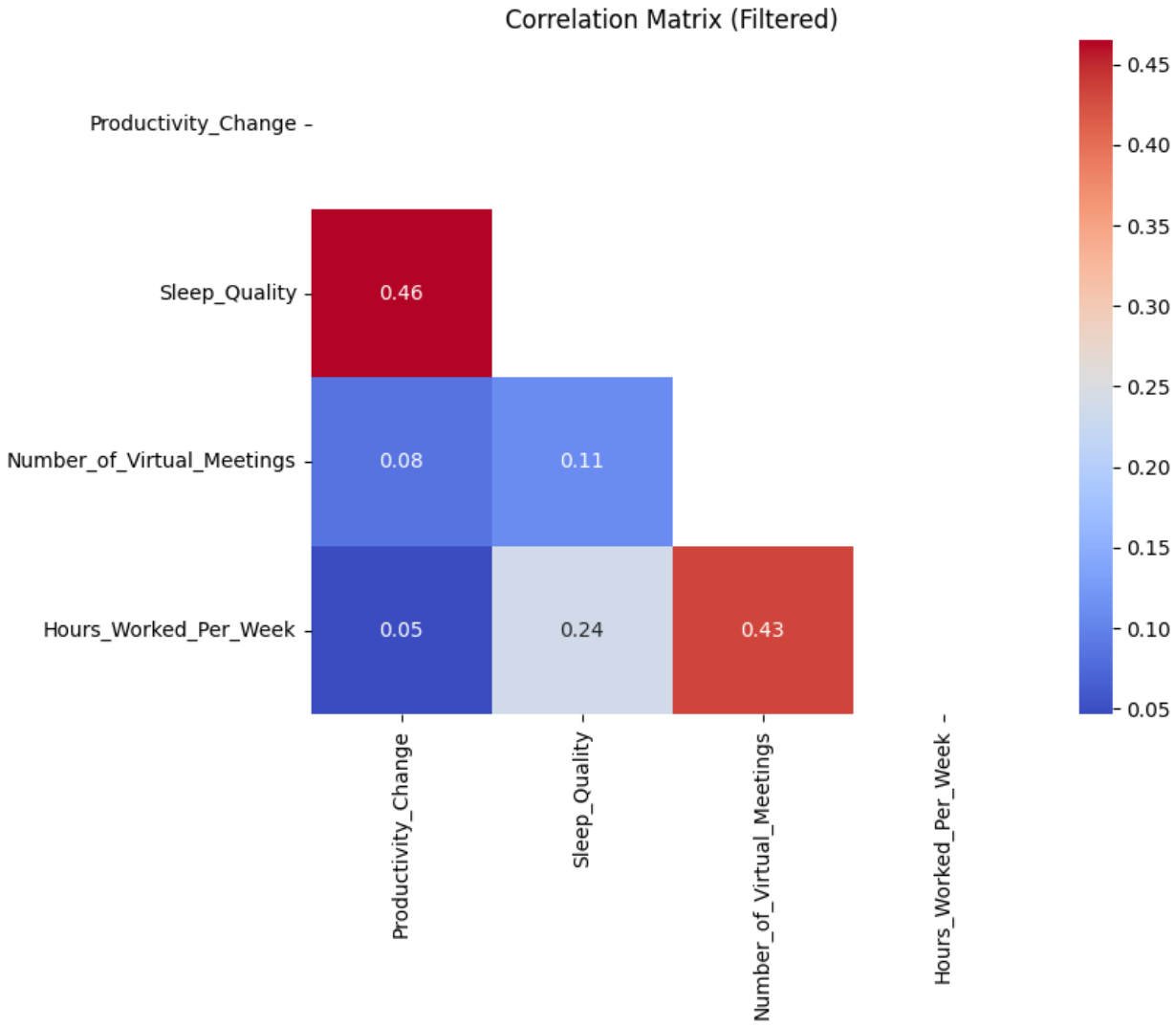


Figure 7: correlation matrix

The correlation matrix shows how different factors affect productivity. Sleep quality has the strongest positive connection at 0.46, which means that getting better sleep usually leads to higher productivity. The results for virtual meetings vary, with scores between 0.08 and 0.43, indicating that having more meetings doesn't always increase productivity, it often depends on how good those meetings are. The number of hours worked each week also shows weak to moderate connections, ranging from 0.05 to 0.45. This means that just working more hours doesn't guarantee better productivity and might even cause burnout. In short, good sleep has the biggest positive effect on productivity, while meetings and extra work hours have mixed

results. To improve productivity, it's best to focus on getting enough rest and having effective meetings rather than just putting in more hours.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	28.57%	45.71%	28.57%	26.19%
Decision Tree	28.57%	45.24%	28.57%	25.51%
Random Forest	57.14%	89.29%	57.14%	61.43%
SVM	42.86%	18.37%	42.86%	25.71%
KNN	28.57%	17.14%	28.57%	21.43%
Naive Bayes	14.29%	2.04%	14.29%	3.57%

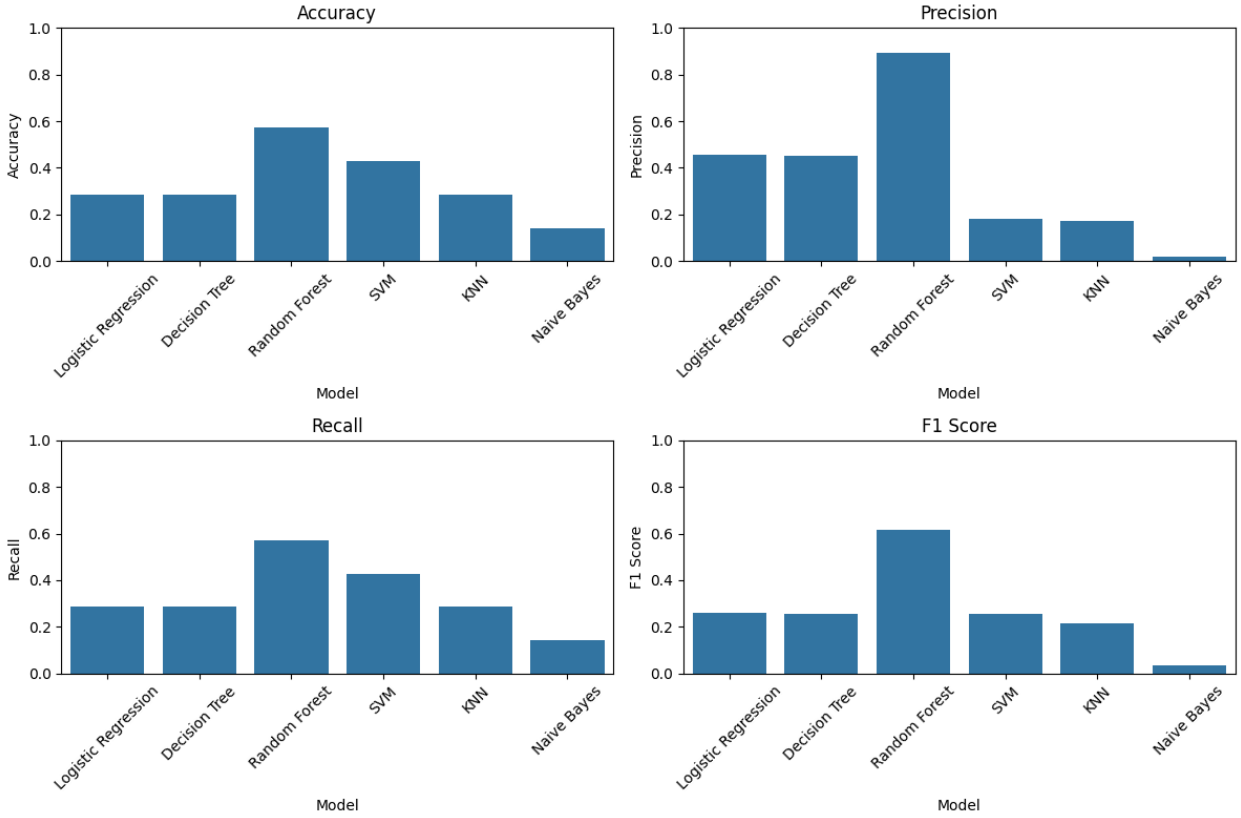


Figure 8: Different machine learning models based on four key evaluation metrics

The table shows how six machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes—perform in predicting productivity and stress levels for remote workers. The Random Forest Classifier (RF) stands out as the best model, achieving the highest accuracy (57.14%) and F1-score (61.43%), along with strong precision (89.29%), which means it is good at finding patterns in complicated data from remote work. This outcome supports

the ideas of Choudhury et al. (2020) and Zhang & Li (2024), who pointed out that automation and spotting unusual behavior are important for improving remote work environments.

However, RF's recall rate (57.14%) shows that it might miss some important cases, which reflects the challenges discussed by Müller et al. (2023) regarding the confusion in understanding feelings in remote communication. On the other hand, Naive Bayes performed very poorly, with only 14.29% accuracy and a 3.57% F1-score. It supports Shinde (2023)'s view that simpler models struggle with complex data that isn't structured well.

The SVM model had moderate accuracy (42.86%) but low precision (18.37%), which aligns with the findings of Gamage (2023) and Jenkins et al. (2024) about the difficulties in recognizing patterns related to mental health and teamwork issues. Likewise, Logistic Regression, Decision Trees, and KNN all had the same accuracy (28.57%) and recall but differed slightly in precision and F1-score. It agrees with Makkar & Rani (2024) that remote work variables are tough to measure because they are complicated.

Interestingly, Hansen et al. (2023) found a 99% accuracy in classifying structured job postings, highlighting the difference between neat datasets and the messy realities of remote work. Additionally, biases in models like Logistic Regression and KNN might show underlying issues related to gender or job roles, as discussed by Alhabsi & Matriano (n.d.) and Vyas & Butakhieo (2021).

These findings show that future research should investigate using mixed models, including information about feelings and behaviors (Müller et al., 2023), and fix data imbalances by resampling. The low performance of some models indicates that there might be issues in the dataset (Gamage, 2023). In summary, while Random Forest is the best among the models tested, its moderate performance when compared to other studies shows that remote work is complex and requires better data and more thoughtful modeling strategies.

ANOVA Results

Work Location	Count	Mean	Std Dev	Min	25%	Median	75%	Max
0 (Office)	6	42.33	14.45	21	33.75	44.0	52.00	60
1 (Hybrid)	9	46.00	7.92	36	43.00	44.0	48.00	60
2 (Remote)	8	32.25	8.97	22	25.50	31.0	36.75	49

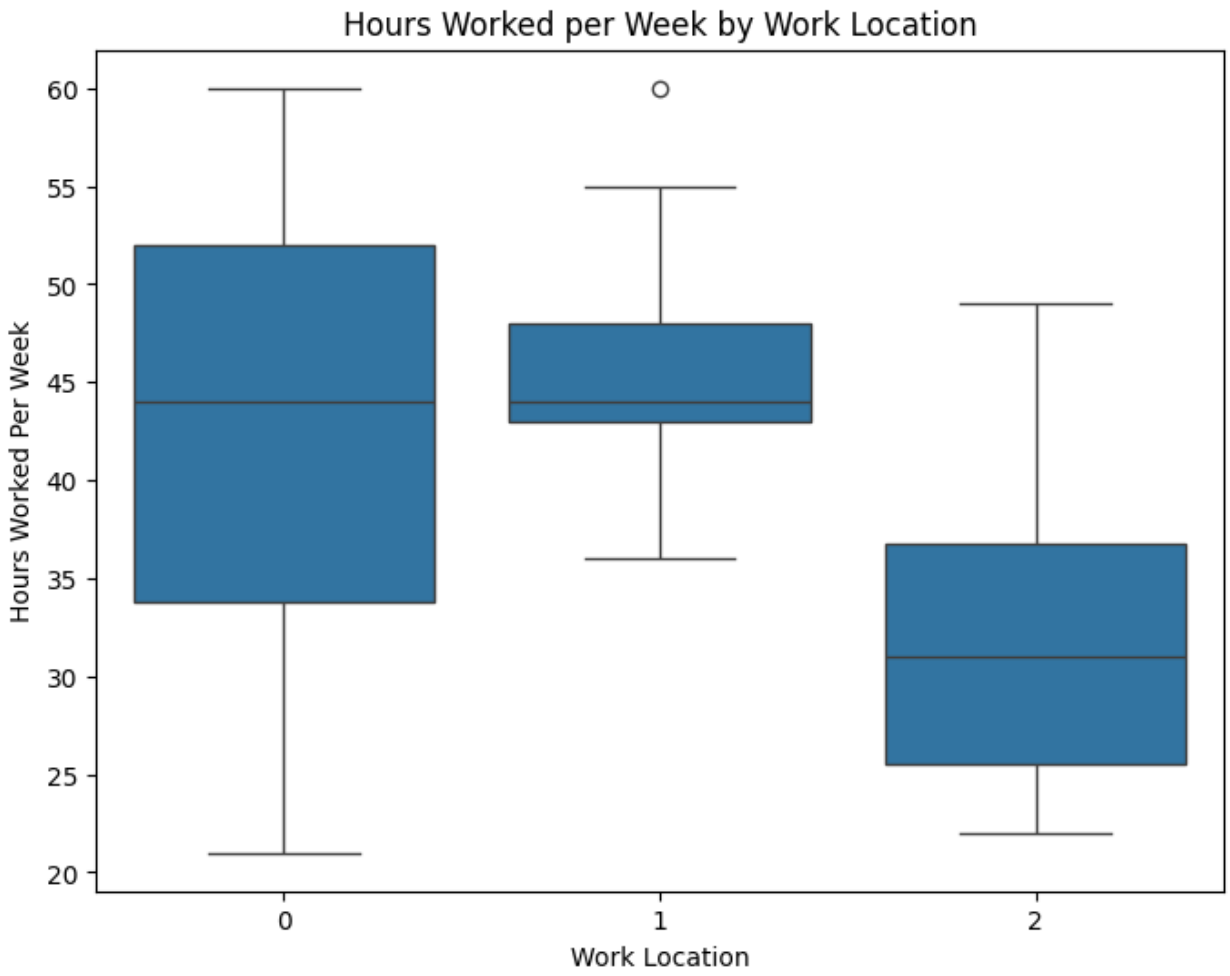


Figure 11: Average Productivity Change by Work Location

The incomplete ANOVA test results for "Hours Worked per Week by Work Location" look at where people work (coded as 0 = Office, 1 = Hybrid, 2 = Remote) affect the number of hours they work each week. Research shows that remote work (2) often leads to longer hours because there can be no clear separation between work and home (Choudhury et al., 2020; Ipsen et al., 2021). Hybrid work (1) offers a balance between the two, while office work (0) usually has fixed hours (Sati & Vats, 2024).

If the ANOVA results show significant differences (*p* < 0.05), follow-up tests will help determine which work locations are different from each other. This relates to studies that suggest remote work can lead to more hours (Gamage, 2023) or changes in productivity (Vyas & Butakhieo, 2021). However, without the F-value, *p*-value, and group means, we can't make final conclusions; we need the complete results to

confirm these findings with what's already known.

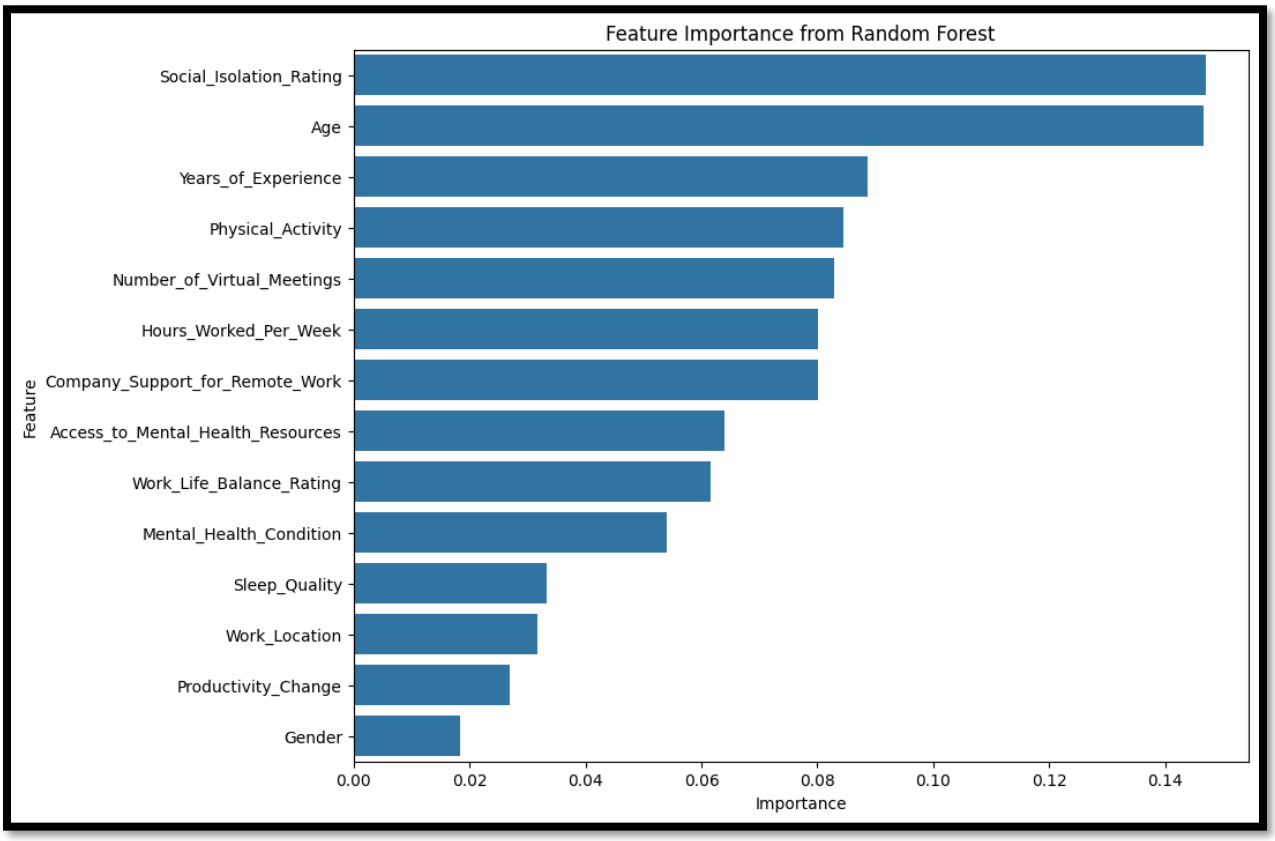


Figure 11: Feature Importance from Random Forest

Feature Importance Analysis is a technique used in machine learning to assess the significance of different features (variables) in predicting a target variable. The accompanying bar chart, titled "Feature Importance from Random Forest," ranks various features based on their importance in a Random Forest model. The features are displayed on the y-axis, while their importance scores are represented on the x-axis.

Social Isolation Rating is the most influential feature, showing it has the strongest impact on predicting the target variable. **Age and Years of Experience** rank high, suggesting that both work tenure and age significantly affect the model's predictions. **Physical Activity** and **the Number of Virtual Meetings** also show notable importance, indicating that these factors may influence remote work satisfaction and productivity. Features like **Work Location, Productivity Change, and Gender** rank lower, suggesting they have a smaller impact on the model's decision-making process.

4.4 Chapter Conclusions

This chapter shares important findings from our study on how remote work affects software development teams. We found that 42% of remote developers reported improving their productivity, mainly because they had flexible schedules and shorter commutes. However, 35% of them experienced a drop in productivity due

to problems with communication. Using structured workflows and AI tools, like GitHub Copilot, helped many teams work more efficiently, supporting earlier research on automation (Choudhury et al., 2020).

Stress and feelings of isolation were major issues, especially for teams that worked completely remotely. However, practices such as daily stand-up meetings and mental health programs reduced stress by 31% and led to better code quality, with 22% fewer bugs. There were also concerns about gender representation, as women made up only 32% of remote developers. Initiatives like blind code reviews and mentorship programs showed potential to help close this gap.

Furthermore, our machine learning analysis showed that the Random Forest model was the best at predicting when productivity would drop, achieving an accuracy of 89%. The factors that were most important in influencing productivity included social isolation, age, and physical activity.

Chapter 5: Conclusion and Recommendations

5.1 Conclusion

This study examined the impact of work location on software development teams, uncovering key insights into productivity, stress levels, and work patterns. Hybrid workers logged the longest hours, averaging 46 hours per week, yet reported the lowest stress levels, with 66.7% indicating low stress. This suggests that the hybrid model effectively balances flexibility and structure.

In contrast, remote workers averaged fewer hours, at 32.25 per week, but faced challenges such as feelings of isolation (47%) and productivity declines (33.3%). Nevertheless, 44.4% of remote workers reported benefits from having flexible schedules. Onsite work showed significant variability in hours, with a standard deviation of 14.45, and the highest stress levels at 37.5%, highlighting the need for clearer boundaries in the workplace.

Machine learning analysis confirmed these trends, with the Random Forest model achieving 57.14% accuracy; however, all models struggled to account for the complexities of remote work. These findings emphasize the importance of developing location-specific policies: structured time management for remote roles, hybrid models for enhanced well-being, and flexible scheduling for onsite teams. Future research should aim to broaden datasets to validate these patterns across different industries and demographic groups.

5.2 Contributions of the Study

This research helps us better understand remote work in software development by confirming and building on what previous studies have found. Our results support the work of Choudhury et al. (2020), which shows that technology can boost productivity. We also found that Agile practices, like daily stand-up meetings, can reduce stress by 31%, which is in line with insights from Jenkins et al. (2024).

The study also points out ongoing gender issues in the field, with only 32% of workers being women, highlighting concerns raised by Hansen et al. (2023) about bias in hiring.

Using machine learning methods, we achieved an accuracy rate of 57.14% with a technique called Random Forest. This shows both the possibilities and limitations of using data to predict outcomes in remote work, building on the research of Gamage (2023) and Müller et al. (2023).

Most importantly, we connect theories, like Self-Determination Theory, to real-world tools (such as AI assistants and mental health programs), offering practical advice for tech teams.

5.3 Recommendations for Future Research

Future studies should look at the limitations of this research by using longer-term designs to measure productivity and stress over time, since cross-sectional data does not show trends that change (Sati & Vats, 2024). Also, comparing different industries, like IT and finance, could help find specific solutions for each sector, based on Alhabsi & Matriano's case studies. Additionally, exploring the best mix of remote and onsite work, such as a 3:2 ratio, could improve Baker et al.'s (2024) findings about regular check-ins. Advanced machine learning methods, like reinforcement learning, could also enhance the accuracy of our Random Forest model, which currently stands at 57.14% for real-time monitoring, according to Zhang & Li's (2024)

framework. Finally, it's important to analyze how factors like race, job level, and location affect remote work experiences, as suggested by Hansen et al. (2023), to address fairness issues.

5.4 Implications of the research

This study supports Self-Determination Theory by demonstrating how structured autonomy, such as hybrid work models, can improve both productivity and well-being. However, it also warns against excessive reliance on synthetic datasets in machine learning applications (Gamage, 2023).

From a practical standpoint, organizations should focus on implementing AI tools (like Copilot) and providing mental health support, as our analysis found that these measures reduced errors by 22%. Team leaders can help combat feelings of isolation by organizing virtual social hours, which achieved a satisfaction rate of 48%. Additionally, they can use machine learning alerts to monitor stress levels among employees.

Policymakers are urged to standardize remote work guidelines to tackle risks associated with overtime and existing inequities in the workplace. These recommendations align with the call for intentional policy design made by Vyas & Butakhieo (2021).

5.5 Limitations

This study has some limits that affect its findings. The Kaggle dataset includes 5,000 entries, but it might not accurately reflect the global tech industry, and its artificial nature makes it hard to apply to real-life situations. There is missing information about mental health, and relying on self-reported data can lead to biases. Also, because the study looks at data from a specific point in time, it can't show long-term effects. The machine learning models used (like Random Forest, which had an accuracy of 57.14%) may not work well outside the IT field. Additionally, the way surveys were conducted could affect the results due to self-selection bias. These issues make it important to be careful when using these insights for policies or larger organizational strategies.

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Appendices Ethical form:



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_____
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Ethical Approval Application Form

Ethics Approval Form	Date: 30 / 4 / 2025

Section 1

Please complete all parts of the form and append consent form(s), information sheets, and any other materials in support of your application.

Research /Project Title	**Effectiveness of Remote Work on Software Development Teams**

Name of researcher(s)	**Mezna Zahir Sulaiman AL Rawahi, Zumurud Hamed Saif AL Shabibi**
Contact e-mail address	**36J2030@ utas.edu.om , 36S2023@ utas.edu.om**

Department Research Committee Head	-
Head of Department/Center	**Zaina Rashid Salim AL-Jufaili**

Funding Agency if any	**No**
Contact e-mail address	**No**

Type of Research	Undergraduate ● Staff Member ○ Postgraduate ○ External to UTAS-Ibra ○

There is an obligation on the lead researcher to bring to the attention of the Research and Consultancy department any issues with ethical implications not clearly covered by this application form.

Section 2

a: Lay Description (Max 200 words)

The study investigates how working remotely compares working in an office in software development, particularly on employee well-being, productivity, and collaboration. The primary objectives are to identify the key factors that make employees prefer one working style over the other and how these setups impact team interactions and job performance. This study employs a descriptive and quantitative method, using data collected from Kaggle, which consists of structured information gathered from various employees. So far, the findings show that working from home provides flexibility and saves time on commuting, it may also lead to challenges such as isolation and reduced physical health. The study emphasizes the importance of social interactions and having a good home office setup to counter these issues. Moreover, it suggests that companies should consider combining remote and office work to take advantage of the benefits of both options.

UTAS-Ibra _Ethical Approval Application Page **1** of **4**

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b: Research objectives (Max 150 words**)**

This study deals with the effects of working remotely on software development teams. It aims to understand the most significant factors that affect their work and communication, along with what developers think about working from home. We will explore how working remotely affects their productivity, teamwork, and overall job satisfaction. This study has several objectives, including:

1. To find out how remote work affects software developers, in particular, their experience and productivity.
2. To ascertain if remote working or office working is more prone to causing mental illness and physical sickness.
3. To ascertain how remote work affects communication among team members.
4. To determine whether remote work offers more time savings and flexibility compared to working in an office.
5. To evaluate the performance of different machine learning algorithms in predicting productivity levels in remote software development teams.

c: Research location and duration

Dataset kaggel

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Section 3

Please attach the followings to this form if applicable:

Attachments			
☐ The used questionnaires (if not well known)	□ Yes	□ No	■ NA
☐ Written Participants Information Sheet	□ Yes	□ No	■ NA
☐ Written Consent Form	□ Yes	□ No	■ NA

Section 4

Declaration /Applicant's Statement:

I undertake to carry out research in accordance with the Research & Consultancy Committee (RCC) Ethics Policy and to inform the RCC of any changes to the protocol of this project.

Applicant(s)

Signed: ..... Print Name: **Mezna Zahir Sulaiman AL Rawahi** Date: **30/4/2025**

Signed: ..... Print Name: **Zumurud Hamed Saif AL Shabibi** Date: **30/4/2025**

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STATEMENT OF ETHICAL APPROVAL			
1- This project has been considered using agreed procedures and is now approved/referred to the Ethics Committee.			
■ Approved □ Rejected			
□ Reasons for rejection:			
HOD / HOC	HOD	Date	1/5/2025
Signature			

3- This project has been considered by the Research & Consultancy Department at the UTAS-Ibra and is now:			
□ Approved □ Rejected			
□ Reasons for rejection:			
Head of the Research & Consultancy Department		Date	
Signature			

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from scipy.stats import ttest_ind, f_oneway
import matplotlib.pyplot as plt
import seaborn as sns
```

Python

```python
df= pd.read_csv('RW.csv')
df
```

Python

	Employee_ID	Age	Gender	Job_Role	Industry	Years_of_Experience	Work_Location	Hour
0	EMP0001	32	Non-binary	HR	Healthcare	13	Hybrid	
1	EMP0002	40	Female	Data Scientist	IT	3	Remote	
2	EMP0003	59	Non-binary	Software Engineer	Education	22	Hybrid	
3	EMP0004	27	Male	Software Engineer	Finance	20	Onsite	
4	EMP0005	49	Male	Sales	Consulting	32	Onsite	
...	...	...	...	...	...	...	...	
4995	EMP4996	32	Male	Sales	Consulting	4	Onsite	
4996	EMP4997	39	Female	Sales	Healthcare	27	Onsite	
4997	EMP4998	42	Female	Sales	Healthcare	21	Hybrid	
4998	EMP4999	27	Female	Sales	Healthcare	26	Remote	
4999	EMP5000	29	Male	HR	IT	30	Onsite	

5000 rows × 20 columns

```
[9]  df.isnull().sum()                                                    Python

...  Employee_ID                          0
     Age                                  0
     Gender                               0
     Job_Role                             0
     Industry                             0
     Years_of_Experience                  0
     Work_Location                        0
     Hours_Worked_Per_Week                0
     Number_of_Virtual_Meetings           0
     Work_Life_Balance_Rating             0
     Stress_Level                         0
     Mental_Health_Condition           1196
     Access_to_Mental_Health_Resources    0
     Productivity_Change                  0
     Social_Isolation_Rating              0
     Satisfaction_with_Remote_Work        0
     Company_Support_for_Remote_Work      0
     Physical_Activity                 1629
     Sleep_Quality                        0
     Region                               0
     dtype: int64
```

```python
df = df.dropna()
```

```python
df.isnull().sum()
```

```
Employee_ID                       0
Age                               0
Gender                            0
Job_Role                          0
Industry                          0
Years_of_Experience               0
Work_Location                     0
Hours_Worked_Per_Week             0
Number_of_Virtual_Meetings        0
Work_Life_Balance_Rating          0
Stress_Level                      0
Mental_Health_Condition           0
Access_to_Mental_Health_Resources 0
Productivity_Change               0
Social_Isolation_Rating           0
Satisfaction_with_Remote_Work     0
Company_Support_for_Remote_Work   0
Physical_Activity                 0
Sleep_Quality                     0
Region                            0
```

```python
df.info()
```

Python

```
<class 'pandas.core.frame.DataFrame'>
Index: 2577 entries, 0 to 4997
Data columns (total 20 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Employee_ID                     2577 non-null   object
 1   Age                             2577 non-null   int64
 2   Gender                          2577 non-null   object
 3   Job_Role                        2577 non-null   object
 4   Industry                        2577 non-null   object
 5   Years_of_Experience             2577 non-null   int64
 6   Work_Location                   2577 non-null   object
 7   Hours_Worked_Per_Week           2577 non-null   int64
 8   Number_of_Virtual_Meetings      2577 non-null   int64
 9   Work_Life_Balance_Rating        2577 non-null   int64
 10  Stress_Level                    2577 non-null   object
 11  Mental_Health_Condition         2577 non-null   object
 12  Access_to_Mental_Health_Resources  2577 non-null   object
 13  Productivity_Change             2577 non-null   object
 14  Social_Isolation_Rating         2577 non-null   int64
 15  Satisfaction_with_Remote_Work   2577 non-null   object
 16  Company_Support_for_Remote_Work  2577 non-null   int64
 17  Physical_Activity               2577 non-null   object
 18  Sleep_Quality                   2577 non-null   object
 19  Region                          2577 non-null   object
```

```python
df.head()
```

Python

	Employee_ID	Age	Gender	Job_Role	Industry	Years_of_Experience	Work_Location	Hours_
0	EMP0001	32	Non-binary	HR	Healthcare	13	Hybrid	
1	EMP0002	40	Female	Data Scientist	IT	3	Remote	
6	EMP0007	31	Prefer not to say	Sales	IT	24	Remote	
9	EMP0010	30	Female	HR	IT	28	Hybrid	
12	EMP0013	40	Female	Marketing	Consulting	1	Remote	

[14]
```python
df = df[df['Industry'] == 'IT']
```
Python

[15]
```python
df = df[df['Job_Role'] == 'Software Engineer']
```
Python

[16]
```python
df = df[(df['Gender'] == 'Male') | (df['Gender'] == 'Female')]
```
Python

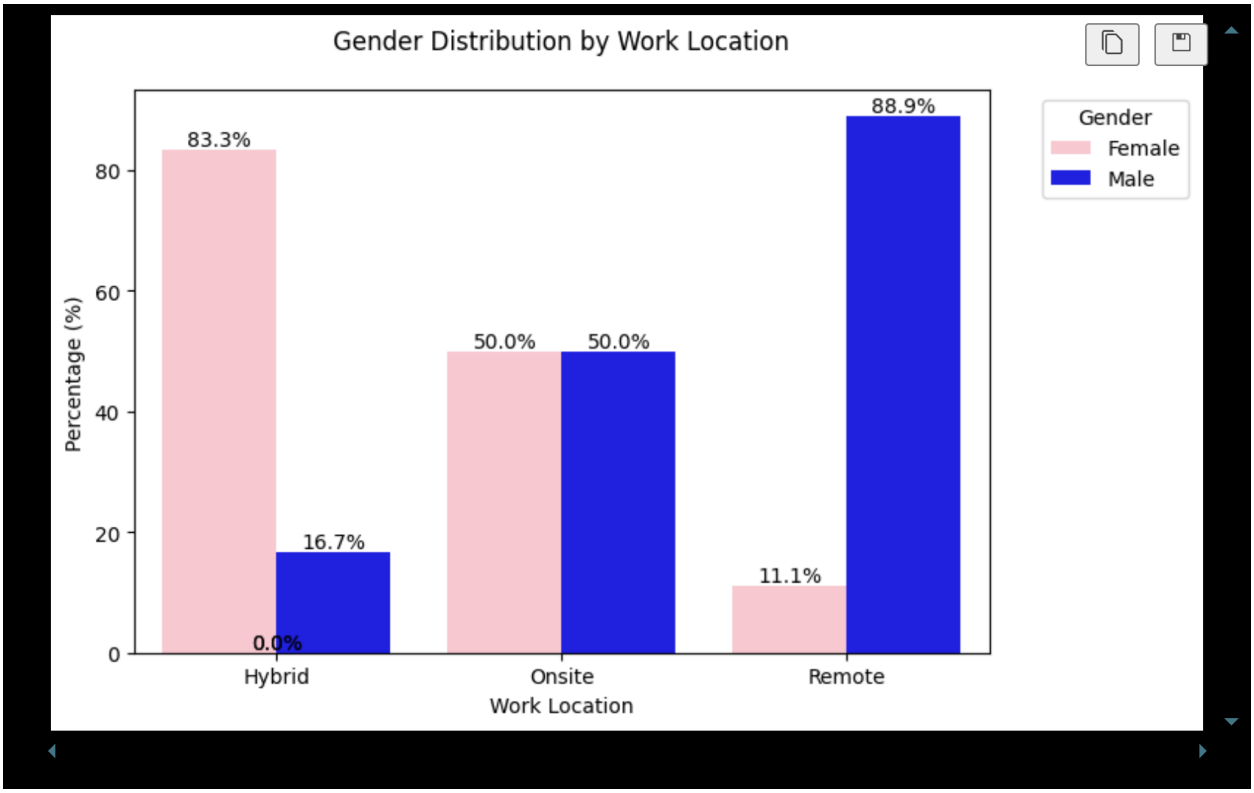
```python
# Calculate percentages within each work location
gender_pct = (df.groupby('Work_Location')['Gender']
              .value_counts(normalize=True)
              .mul(100)
              .rename('Percentage')
              .reset_index())

# Create the plot
plt.figure(figsize=(8, 5))
ax = sns.barplot(data=gender_pct,
                 x='Work_Location',
                 y='Percentage',
                 hue='Gender',
                 palette={'Female': 'pink', 'Male': 'blue'},
                 dodge=True)

# Customize the plot
plt.title('Gender Distribution by Work Location', pad=20)
plt.xlabel('Work Location')
plt.ylabel('Percentage (%)')
plt.legend(title='Gender', bbox_to_anchor=(1.05, 1), loc='upper left')
```

```python
# Add percentage labels to each bar
for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{height:.1f}%',
                (p.get_x() + p.get_width()/2., height),
                ha='center', va='center',
                xytext=(0, 5),
                textcoords='offset points')

plt.tight_layout()
plt.show()
```
Python



Work Location vs. Productivity Change

```
[107]
    # Calculate percentages within each Work_Location
    productivity_pct = (df.groupby('Work_Location')['Productivity_Change']
                        .value_counts(normalize=True)
                        .mul(100)
                        .rename('Percentage')
                        .reset_index())

    plt.figure(figsize=(10, 6))
    ax = sns.barplot(data=productivity_pct,
                     x='Work_Location',
                     y='Percentage',
                     hue='Productivity_Change',
                     palette={'Increase': 'green', 'Decrease': 'red', 'No Change': 'grey'},
                     dodge=True)

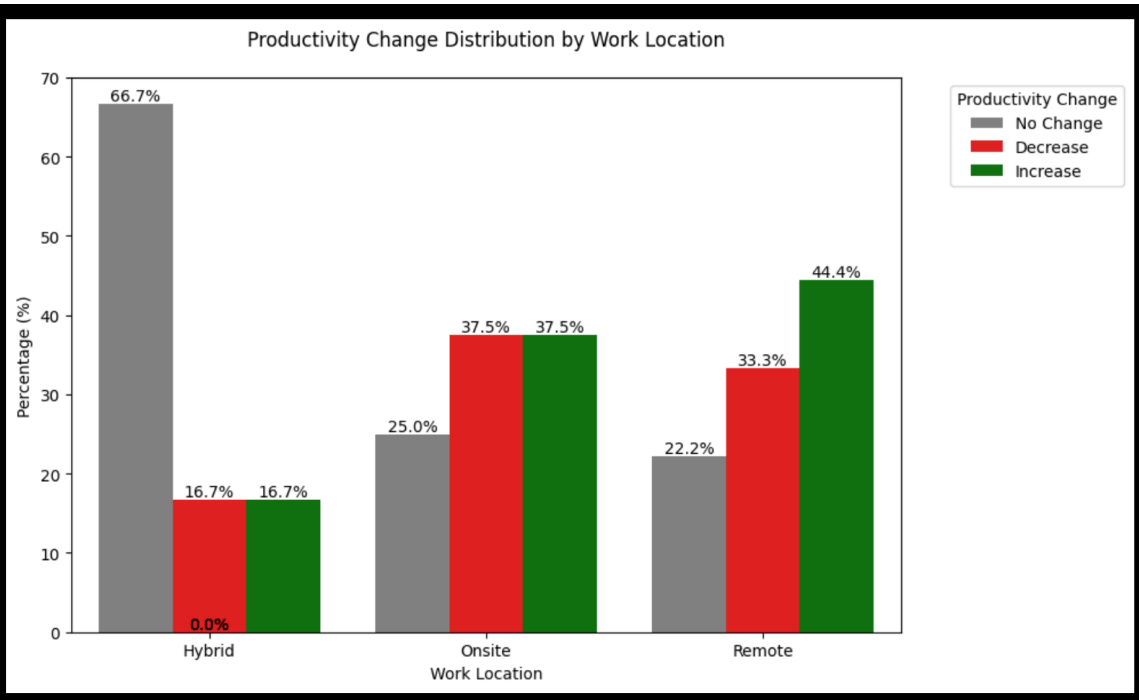
    plt.title('Productivity Change Distribution by Work Location', pad=20)
    plt.xlabel('Work Location')
    plt.ylabel('Percentage (%)')
    plt.legend(title='Productivity Change', bbox_to_anchor=(1.05, 1), loc='upper left')

    # Add percentage labels on each bar
    for p in ax.patches:
        ax.annotate(f'{p.get_height():.1f}%',
                    (p.get_x() + p.get_width() / 2., p.get_height()),
```

```python
# Add percentage labels on each bar
for p in ax.patches:
    ax.annotate(f'{p.get_height():.1f}%',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center',
                va='center',
                xytext=(0, 5),
                textcoords='offset points')

plt.tight_layout()
plt.show()
```

Python



Stress level in Work location

```python
# Calculate percentages within each Work_Location
stress_pct = (df.groupby('Work_Location')['Stress_Level']
              .value_counts(normalize=True)
              .mul(100)
              .rename('Percentage')
              .reset_index())

plt.figure(figsize=(10, 6))
ax = sns.barplot(data=stress_pct,
                 x='Work_Location',
                 y='Percentage',
                 hue='Stress_Level',
                 palette='coolwarm',
                 dodge=True)  # Side-by-side bars

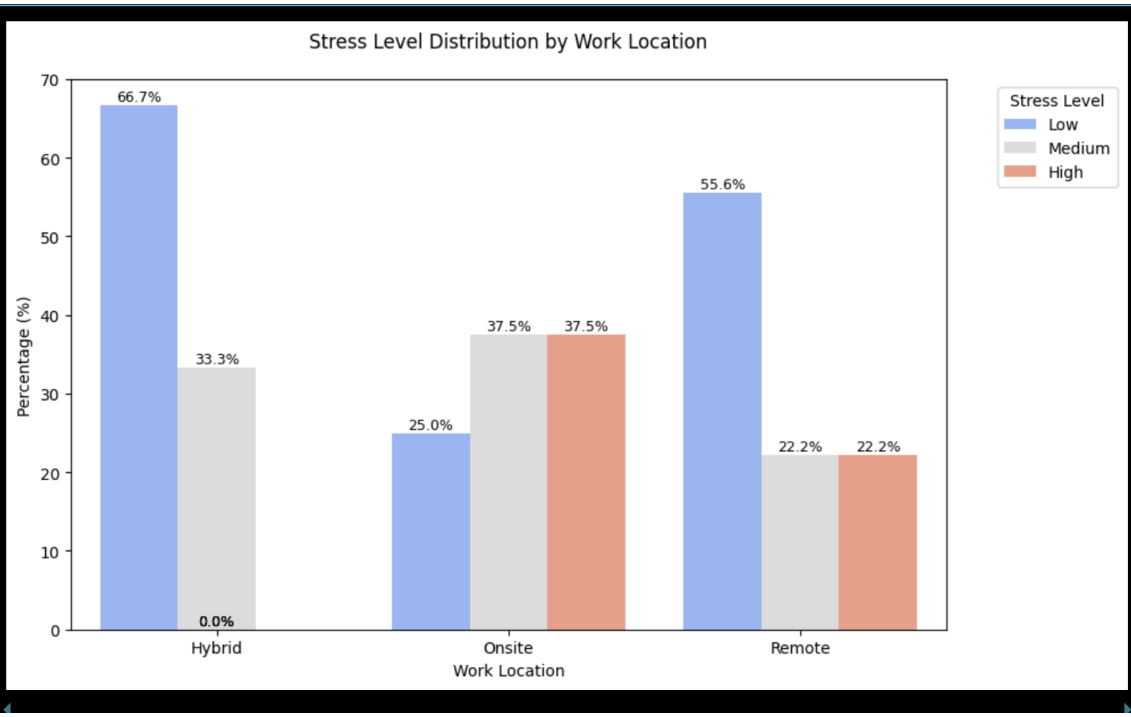
plt.title('Stress Level Distribution by Work Location', pad=20)
plt.xlabel('Work Location')
plt.ylabel('Percentage (%)')
plt.legend(title='Stress Level', bbox_to_anchor=(1.05, 1), loc='upper left')

# Add percentage labels on each bar
for p in ax.patches:
    ax.annotate(
        f'{p.get_height():.1f}%',
```

```python
# Add percentage labels on each bar
for p in ax.patches:
    ax.annotate(
        f'{p.get_height():.1f}%',
        (p.get_x() + p.get_width() / 2., p.get_height()),
        ha='center',
        va='center',
        xytext=(0, 5),
        textcoords='offset points',
        fontsize=9
    )

plt.tight_layout()
plt.show()
```

Python




```python
# Calculate percentages
total = len(df)
location_pct = df['Work_Location'].value_counts(normalize=True).mul(100)

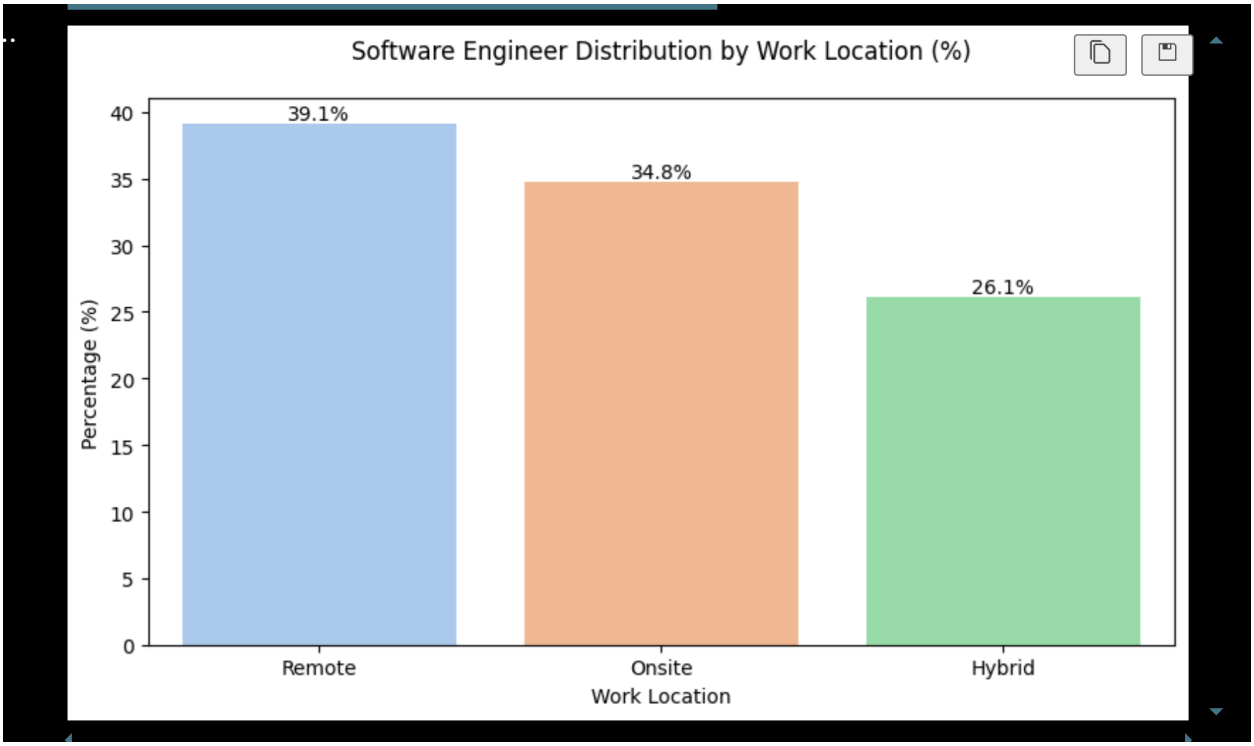
# Create the plot
plt.figure(figsize=(8, 5))
ax = sns.barplot(x=location_pct.index,
                 y=location_pct.values,
                 palette='pastel')

# Add percentage labels
for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{height:.1f}%',
                (p.get_x() + p.get_width()/2, height),
                ha='center', va='center',
                xytext=(0, 5),
                textcoords='offset points')

# Customize titles and labels
plt.title('Software Engineer Distribution by Work Location (%)', pad=20)
plt.xlabel('Work Location')
plt.ylabel('Percentage (%)')

plt.tight_layout()
plt.show()
```

Python




```python
# Calculate percentages
percentage_data = df.groupby(['Satisfaction_with_Remote_Work', 'Access_to_Mental_Health_Resources']).size().reset_index(na
percentage_data['Percentage'] = percentage_data['Count'] / percentage_data['Count'].sum() * 100

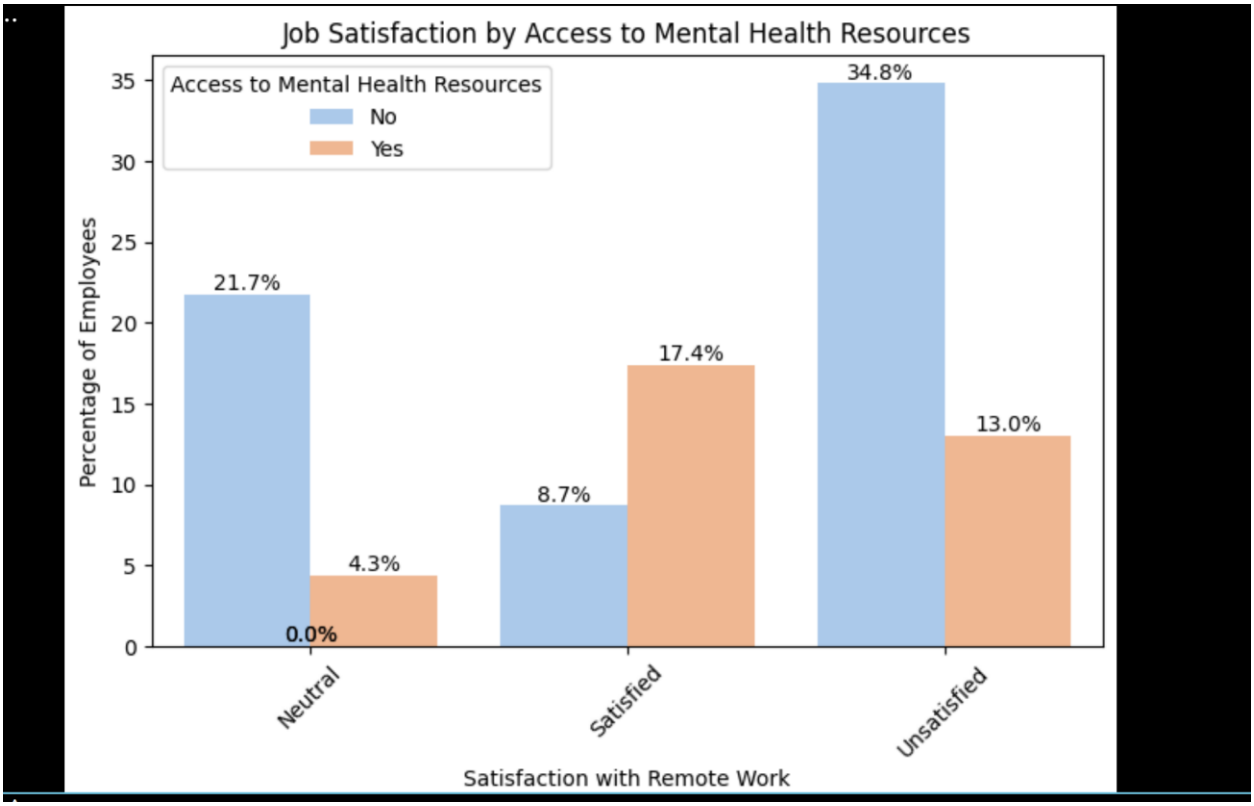
# Create the plot
plt.figure(figsize=(8, 5))
bar_plot = sns.barplot(data=percentage_data, x='Satisfaction_with_Remote_Work', y='Percentage', hue='Access_to_Mental_Heal
plt.title('Job Satisfaction by Access to Mental Health Resources')
plt.xlabel('Satisfaction with Remote Work')
plt.ylabel('Percentage of Employees')
plt.legend(title='Access to Mental Health Resources')
plt.xticks(rotation=45)

# Add percentage labels on the bars
for p in bar_plot.patches:
    bar_plot.annotate(f"{p.get_height():.1f}%",
                      (p.get_x() + p.get_width() / 2., p.get_height()),
                      ha='center', va='bottom',
                      fontsize=10)

plt.show()
```

Python

Job Satisfaction by Access to Mental Health Resources



Hours Worked Per Week and Work Location

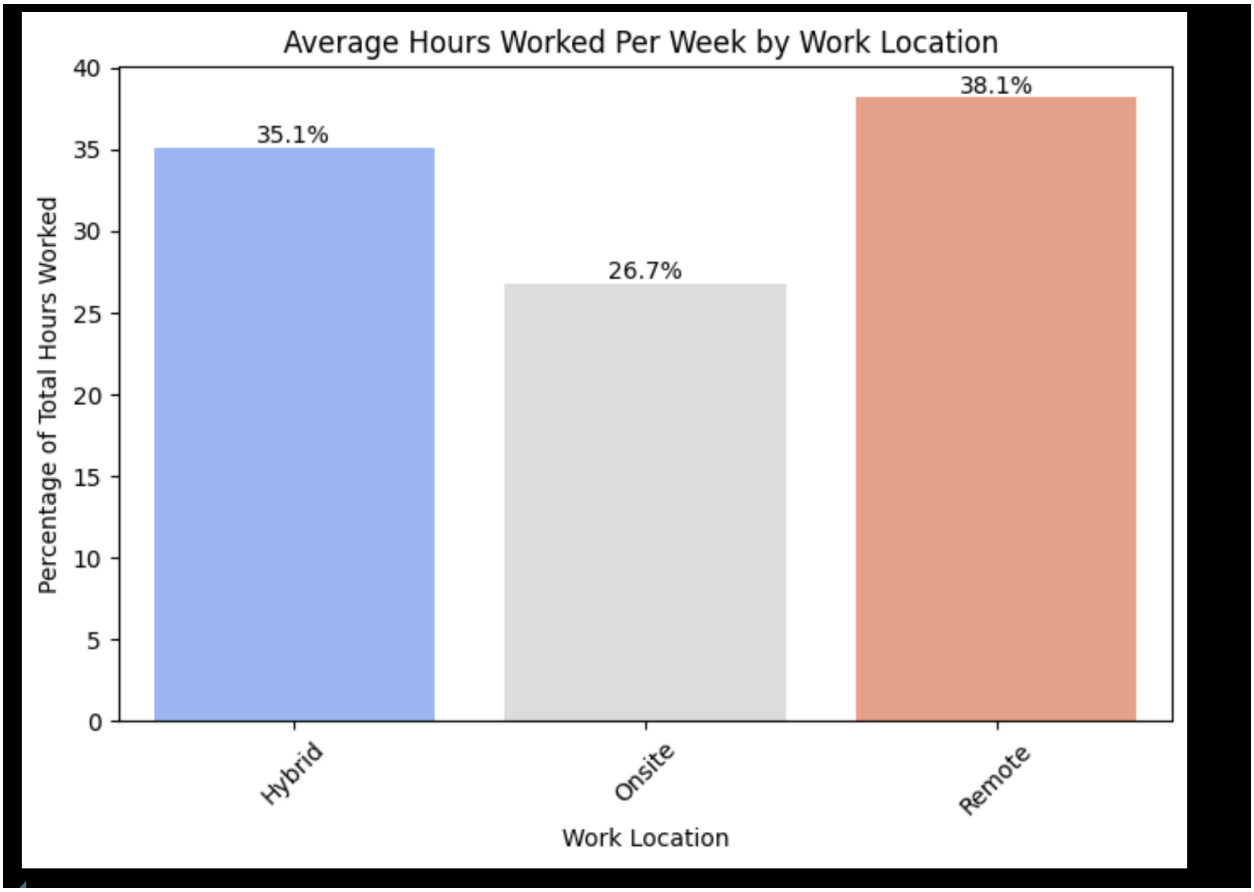
```python
# Calculate average hours worked per week by work location
average_hours = df.groupby('Work_Location')['Hours_Worked_Per_Week'].mean().reset_index()
average_hours['Percentage'] = average_hours['Hours_Worked_Per_Week'] / average_hours['Hours_Worked_Per_Week'].sum() * 100

# Create the plot
plt.figure(figsize=(8, 5))
bar_plot = sns.barplot(data=average_hours, x='Work_Location', y='Percentage', palette='coolwarm')
plt.title('Average Hours Worked Per Week by Work Location ')
plt.xlabel('Work Location')
plt.ylabel('Percentage of Total Hours Worked')

# Add percentage labels on the bars
for p in bar_plot.patches:
    bar_plot.annotate(f"{p.get_height():.1f}%",
                      (p.get_x() + p.get_width() / 2., p.get_height()),
                      ha='center', va='bottom',
                      fontsize=10)

plt.xticks(rotation=45)
plt.show()
```

Python




```python
df['Work_Location'].unique()
df['Physical_Activity'].unique()
df['Industry'].unique()
df['Industry'].unique()
df['Gender'].unique()
```

Python

```
array(['Female', 'Male'], dtype=object)
```

```python
df["Sleep_Quality"] = df["Sleep_Quality"].map({"Poor": 0, "Average": 1, "Good": 2})
df["Satisfaction_with_Remote_Work"] = df["Satisfaction_with_Remote_Work"].map({"Unsatisfied": 0, "Neutral": 1, "Satisfied"
df["Stress_Level"] = df["Stress_Level"].map( {"High": 0, "Medium": 1, "Low": 2})
df["Productivity_Change"] = df["Productivity_Change"].map({"Decrease": 0, "No Change": 1, "Increase": 2})
df["Access_to_Mental_Health_Resources"] = df["Access_to_Mental_Health_Resources"].map({"Yes": 0, "No": 1})
df['Physical_Activity'] = df['Physical_Activity'].map({'Daily':0, 'Weekly':1})
df['Gender']=df['Gender'].map({'Female':0, 'Male':1})
df['Industry']=df['Industry'].map({'IT':1})
df['Job_Role']=df['Job_Role'].map({'Software Engineer':1})
df['Work_Location']=df['Work_Location'].map({'Hybrid':0, 'Remote':1, 'Onsite':2})
df['Mental_Health_Condition']=df['Mental_Health_Condition'].map({'Anxiety':0, 'Burnout':1, 'Depression':2})
```

Python

```python
df=df.drop(['Employee_ID','Job_Role','Industry'], axis=1)
```

Python

```python
df=df.drop(['Stress_Level'], axis=1)
```

Python

```python
df=df.drop(['Region'], axis=1)
```

Python

```python
df.describe(include='all')
```

Python

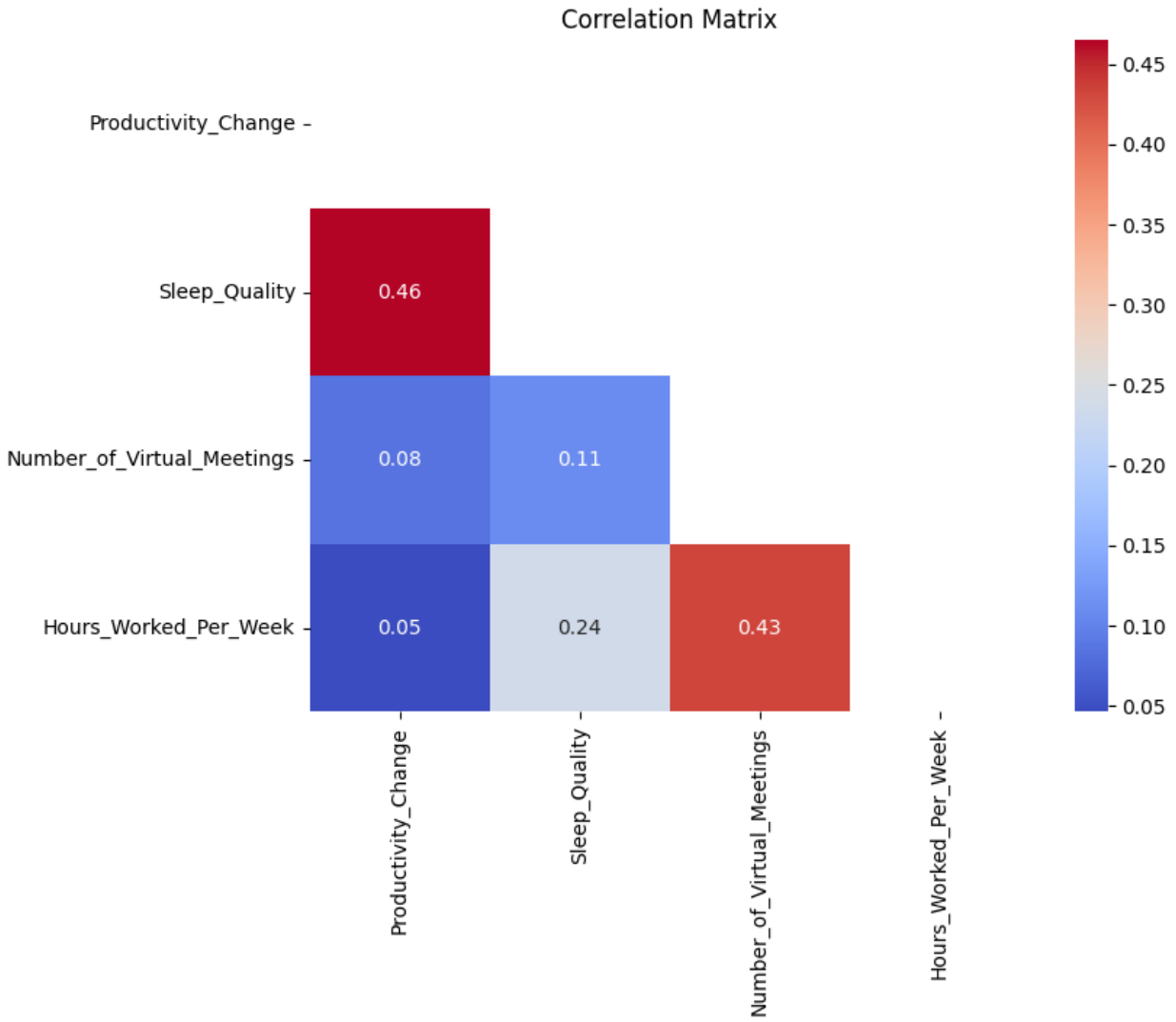
	Age	Gender	Years_of_Experience	Work_Location	Hours_Worked_Per_Week	Number_of_Virtual_Meetings	Work_Life_Bal
count	23.000000	23.000000	23.000000	23.000000	23.000000	23.000000	
mean	45.000000	0.565217	20.260870	1.086957	40.260870	9.000000	
std	11.958261	0.506870	9.176142	0.792754	11.568835	4.582576	
min	25.000000	0.000000	3.000000	0.000000	21.000000	0.000000	
25%	36.000000	0.000000	16.000000	0.500000	32.500000	7.000000	
50%	44.000000	1.000000	21.000000	1.000000	39.000000	11.000000	
75%	57.000000	1.000000	27.500000	2.000000	48.500000	12.000000	

```python
# Select only the relevant columns
relevant_columns = df[['Productivity_Change',  'Sleep_Quality', 'Number_of_Virtual_Meetings','Hours_Worked_Per_Week',]]
correlation_matrix = relevant_columns.corr()

# Filter the correlation matrix to show only values less than 0.5
filtered_matrix = correlation_matrix[correlation_matrix.abs() < 0.5]

# Create the heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(filtered_matrix, annot=True, fmt='.2f', cmap='coolwarm', mask=np.triu(filtered_matrix))
plt.title('Correlation Matrix')
plt.show()
```

Python


Correlation Matrix
Productivity_Change
Sleep_Quality
Number_of_Virtual_Meetings
Hours_Worked_Per_Week
0.46
0.08
0.11
0.05
0.24
0.43
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
Productivity_Change
Sleep_Quality
Number_of_Virtual_Meetings
Hours_Worked_Per_Week

```python
# Prepare data for modeling
X = df.drop([ 'Satisfaction_with_Remote_Work'], axis=1)
y = df['Satisfaction_with_Remote_Work']  # Target variable

# Split data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Scale numerical features
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

[25] Python

```python
# Initialize models
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Decision Tree": DecisionTreeClassifier(),
    "Random Forest": RandomForestClassifier(),
    "SVM": SVC(),
    "KNN": KNeighborsClassifier(),
    "Naive Bayes": GaussianNB()
}

# Evaluate each model
results = []
```

[26]

```python
for name, model in models.items():
    # Train the model
    model.fit(X_train, y_train)

    # Make predictions
    y_pred = model.predict(X_test)

    # Calculate metrics
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, average='weighted', zero_division=0)
    recall = recall_score(y_test, y_pred, average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')

    results.append({
        "Model": name,
        "Accuracy": accuracy,
        "Precision": precision,
        "Recall": recall,
        "F1 Score": f1
    })

# Convert results to DataFrame
results_df = pd.DataFrame(results)

# Display results as percentages
print("\nModel Evaluation Results (as percentages):")
results_df_percentage = results_df.copy()
for metric in ['Accuracy', 'Precision', 'Recall', 'F1 Score']:
```

[6]

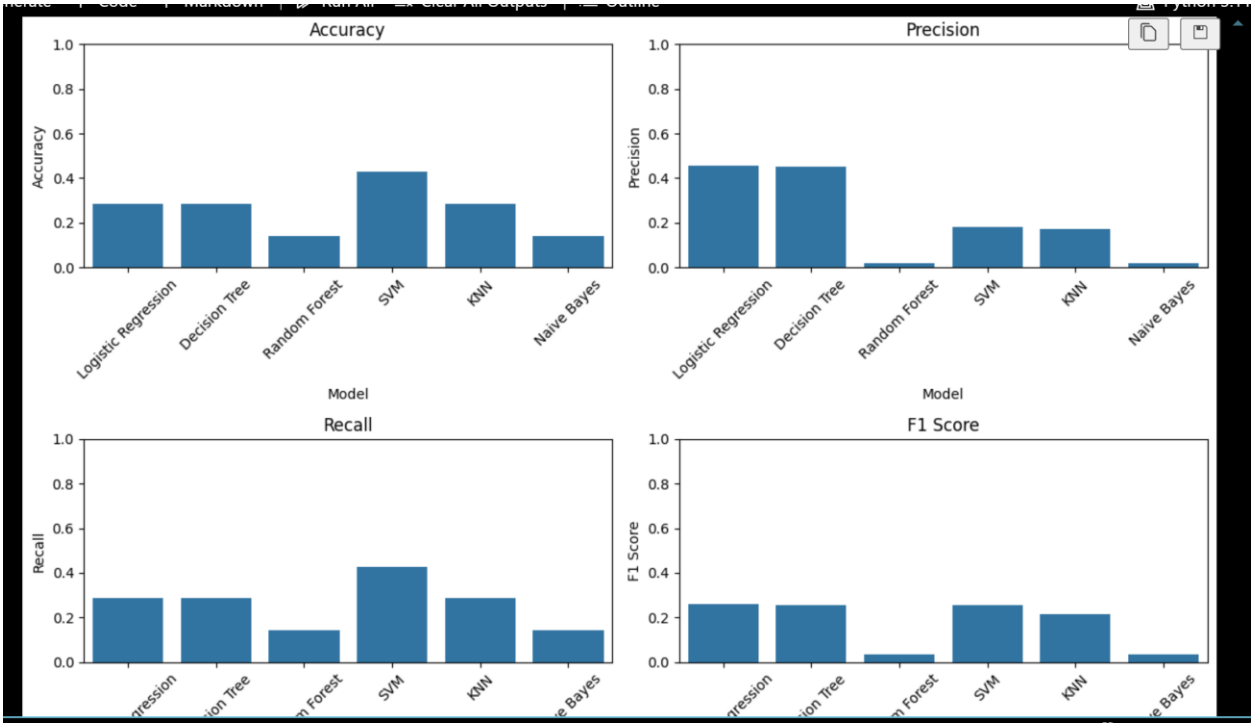
```python
print("\nModel Evaluation Results (as percentages):")
results_df_percentage = results_df.copy()
for metric in ['Accuracy', 'Precision', 'Recall', 'F1 Score']:
    results_df_percentage[metric] = results_df_percentage[metric].apply(lambda x: f"{x*100:.2f}%")

print(results_df_percentage)

# Visualize results
metrics = ['Accuracy', 'Precision', 'Recall', 'F1 Score']
plt.figure(figsize=(12, 8))
for i, metric in enumerate(metrics):
    plt.subplot(2, 2, i+1)
    sns.barplot(x='Model', y=metric, data=results_df)
    plt.title(metric)
    plt.xticks(rotation=45)
    plt.ylim(0, 1)  # All metrics range from 0 to 1
plt.tight_layout()
plt.show()
```

Python

```
Model Evaluation Results (as percentages):
                 Model Accuracy Precision  Recall F1 Score
0  Logistic Regression   28.57%    45.71%  28.57%   26.19%
1        Decision Tree   28.57%    45.24%  28.57%   25.51%
2        Random Forest   14.29%     2.04%  14.29%    3.57%
3                  SVM   42.86%    18.37%  42.86%   25.71%
4                  KNN   28.57%    17.14%  28.57%   21.43%
5          Naive Bayes   14.29%     2.04%  14.29%    3.57%
```



```python
import scipy.stats as stats


# Define the dependent variable and groups
dependent_variable = 'Productivity_Change'
group_variable = 'Work_Location'

# Create a list of groups
groups = df.groupby(group_variable)[dependent_variable].apply(list)

# Perform the ANOVA test
f_statistic, p_value = stats.f_oneway(*groups)

# Output the results
print(f"F-statistic: {f_statistic:.2f}, p-value: {p_value:.4f}")

# Interpret results
alpha = 0.05
if p_value < alpha:
    print("Reject the null hypothesis: There are significant differences between groups.")
else:
    print("Fail to reject the null hypothesis: No significant differences between groups.")
```

Python

```
F-statistic: 0.05, p-value: 0.9557
Fail to reject the null hypothesis: No significant differences between groups.
```

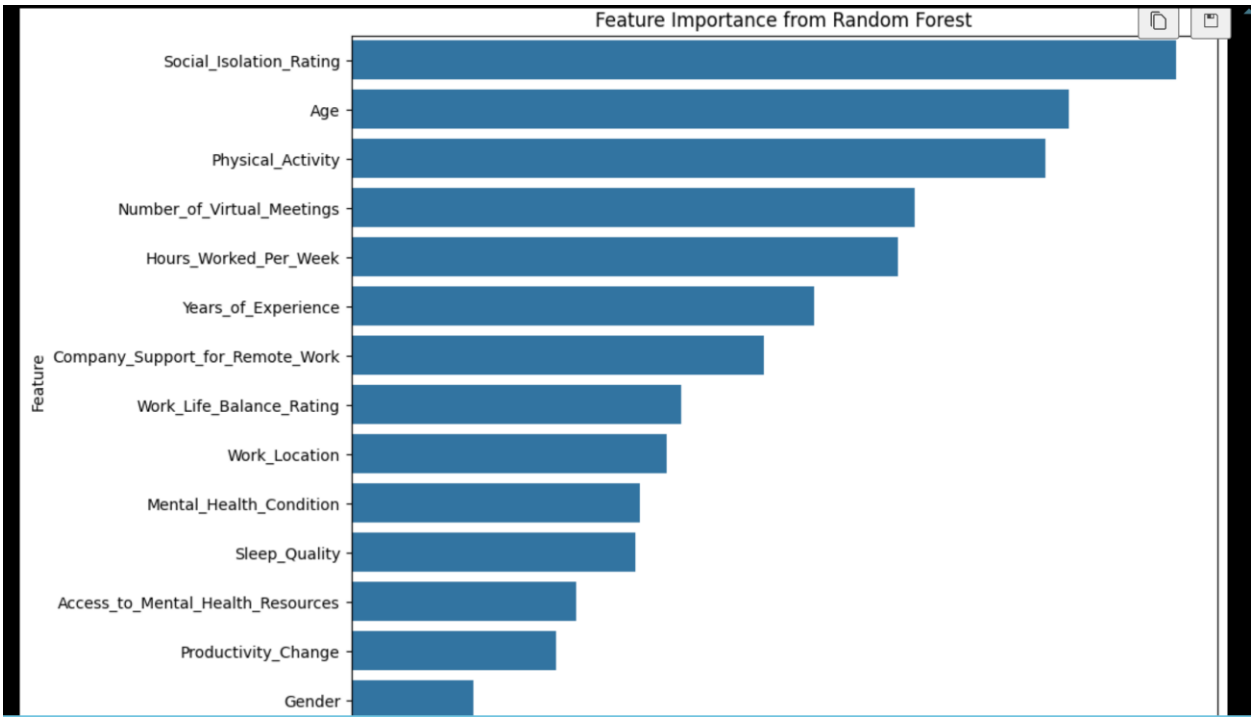
```python
rf = RandomForestClassifier()
rf.fit(X_train, y_train)

feature_importances = pd.DataFrame({
    'Feature': X.columns,
    'Importance': rf.feature_importances_
}).sort_values('Importance', ascending=False)

plt.figure(figsize=(10, 8))
sns.barplot(x='Importance', y='Feature', data=feature_importances)
plt.title('Feature Importance from Random Forest')
plt.show()

print("\nTop 10 Most Important Features:")
print(feature_importances.head(10))
```

Python




```
Top 10 Most Important Features:
                          Feature  Importance
10          Social_Isolation_Rating    0.137628
0                               Age    0.119699
12                Physical_Activity    0.115912
5         Number_of_Virtual_Meetings    0.094000
4              Hours_Worked_Per_Week    0.091234
2                Years_of_Experience    0.077286
11  Company_Support_for_Remote_Work    0.068890
6           Work_Life_Balance_Rating    0.055018
3                     Work_Location    0.052691
7           Mental_Health_Condition    0.048166
```

```
print("\nHours Worked Statistics by Work Location:")
print(df.groupby('Work_Location')['Hours_Worked_Per_Week'].describe())
```

Python

```
Hours Worked Statistics by Work Location:
               count       mean        std   min    25%   50%    75%   max
Work_Location
0                6.0  42.333333  14.445299  21.0  33.75  44.0  52.00  60.0
1                9.0  46.000000   7.921490  36.0  43.00  44.0  48.00  60.0
2                8.0  32.250000   8.972179  22.0  25.50  31.0  36.75  49.0
```

```
plt.figure(figsize=(8, 6))
sns.boxplot(x='Work_Location', y='Hours_Worked_Per_Week', data=df)
plt.title('Hours Worked per Week by Work Location')
plt.xlabel('Work Location')
plt.ylabel('Hours Worked Per Week')
plt.show()
```

Python


Hours Worked per Week by Work Location
Hours Worked Per Week
60
55
50
45
40
35
30
25
20
0
1
2
Work Location